# Kazhdan-Lusztig Theory

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# 1 Hecke Algebra

Let's start by taking a very classical point of view. Let $X$ be a finite set, we have a convolution algebra: $(\mathrm{Fun}(X \times X), *)$ given by:

$$(f * g)(x, y) = \sum_{z \in X} f(x, z) g(z, y)$$

This algebra has a unit $\mathbf{1}_{\Delta_X}$. It is clear that:

$$(\mathrm{Fun}(X \times X), *) \simeq \mathrm{End}_{\mathbb{C}}(\mathrm{Fun}(X))$$

Let $G$ act on $X$, by putting $G$-invariant restriction we have identification:

$$(\mathrm{Fun}(X \times X)^G, *) \simeq \mathrm{End}_G(\mathrm{Fun}(X))$$

Apply this construction to $G = \underline{G}(\mathbb{F}_q), X = \mathcal{B}(\mathbb{F}_q)$, where $\underline{G}$ is some split reductive group over $\mathbb{F}_q$ with flag variety $\mathcal{B}$. We have:

**Definition 1.1.** The Hecke algebra $\mathcal{H}_G = \mathrm{Fun}(\mathcal{B}(\mathbb{F}_q) \times \mathcal{B}(\mathbb{F}_q))^{\underline{G}(\mathbb{F}_q)}$.

Note we have isomorphisms:

$$\mathcal{H}_G \simeq \mathrm{End}_{\underline{G}(\mathbb{F}_q)}(\mathrm{Fun}(\mathcal{B}(\mathbb{F}_q))) \simeq \mathrm{End}_{\underline{G}(\mathbb{F}_q)}(\mathrm{Ind}_{\underline{B}(\mathbb{F}_q)}^{\underline{G}(\mathbb{F}_q)} \mathbb{C})$$

**Example.** *Consider the easiest case with $G = \mathrm{SL}_2(\mathbb{F}_q), X = \mathbb{P}^1(\mathbb{F}_q)$. Note that we have $G$ orbit decomposition:*

$$X \times X = \Delta X \coprod (X \times X \backslash \Delta X)$$

*We define $T_1 = \mathbf{1}_{\Delta X}, T_s = \mathbf{1}_{X \times X \backslash \Delta X}$. These form a basis for $\mathcal{H}_G$. We only need to compute for $\ell, \ell' \in \mathbb{P}^1(\mathbb{F}_q)$:*

$$T_s^2(\ell, \ell') = \sum_{\ell'' \in \mathbb{P}^1(\mathbb{F}_q)} T_s(\ell, \ell'') T_s(\ell'', \ell') = \begin{cases} q, & \text{if } \ell = \ell'; \\ q - 1, & \text{otherwise.} \end{cases}$$

*We thus have:* $T_s^2 = qT_1 + (q-1)T_s \iff (T_s + 1)(T_s - q) = 0$

In general, the orbits are parametrized by $W$ and we the orbit decomposition is:

$$X \times X = \coprod_{w \in W} \mathcal{O}(w)(\mathbb{F}_q)$$

Hence $\mathcal{H}_G$ has a basis: $\{T_w = \mathbf{1}_{\mathcal{O}(w)(\mathbb{F}_q)}\}_{w \in W}$. Hecke algebra $\mathcal{H}_G$ is generated by $T_s, s \in S$ (corresponding to simple reflections) by two kinds of relations:

1. Quadratic relations. For all $s$, $(T_s + 1)(T_s - q) = 0$. To prove this, pick standard parabolic $P_s \subset G$ corresponding to $s$ with Levi quotient $L_s$. We have $\mathcal{B}_{L_s} \simeq \mathcal{B}_{P_s} \subset \mathcal{B}$. Then the extension-by-zero induces an algebra homomorphism $\mathcal{H}_{L_s} \subset \mathcal{H}_G$. But $T_s \in \mathcal{H}_{L_s} \simeq \mathcal{H}_{\mathrm{PGL}_2}$, so the relation is simply the $\mathrm{SL}_2$ computation we did before.

2. Braid relation. For $s \neq t \in S$, we have

$$\underbrace{T_s T_t \cdots}_{m_{s,t}} = \underbrace{T_t T_s \cdots}_{m_{s,t}}$$

This follows from the braid relations in Weyl group providing we can prove

**Lemma 1.2.** *For a reduced expression $w = s_1 \cdots s_l \in W$, $T_w = T_{s_1} \cdots T_{s_l}$*

The proof of this lemma is more or less the definition of the convolution product as well as the fact that each pair of flags $x_0, x_l \in \mathcal{B}(\mathbb{F}_q)$ in relative position $w$ can be uniquely filled into a chain $(x_0, \cdots, x_l)$ with $(x_i, x_{i+1})$ has relative position $s_i$ for each $i$.

**Lemma 1.3.** *The ring homomorphism:*

$$\mathbb{C}\langle T_s \mid s \in S\rangle/(\text{relations above}) \to \mathcal{H}_G$$

*is an isomorphism.*

*Proof.* Surjectivity is clear. To show injectivity we only neeed to bound the dimension of LHS. This is to show for any sequence $(s_i \in S)_{1 \leq i \leq n}$ :

$$T_{s_1} \cdots T_{s_n} \in \text{Span}(T_w \mid w \in W)$$

but this is clear by reducing the length of the word $s_1 \cdots s_n$. □

**Definition 1.4.** The generic Hecke algebra over base $\mathbb{Z}[v, v^{-1}]$ is

$$\mathcal{H}_v(W) \coloneqq \mathbb{Z}[v, v^{-1}]\langle T_s \mid s \in S\rangle/((T_s + 1)(T_s - v^2), \text{briad rel})$$

So that we see:

$$\mathcal{H}_v(W) \otimes_{\mathbb{Z}[v,v^{-1}]} \mathbb{C} \simeq \mathcal{H}_G$$

for $v \mapsto \pm q^{\frac{1}{2}}$, and

$$\mathcal{H}_v(W) \otimes_{\mathbb{Z}[v,v^{-1}]} \mathbb{Z} \simeq \mathbb{Z}[W]$$

for $v \mapsto \pm 1$.

There is a distinguished involution:

$$\overline{\bullet} : \mathcal{H}_v(W) \to \mathcal{H}_v(W)$$

by $\overline{T}_s = T_s^{-1}, \overline{v} = v^{-1}$. It is clear that $\overline{T_w} = T_{w^{-1}}^{-1}$.

**Remark.** *An important calculation is*

$$T_s^{-1} = v^{-2} T_s - (1 - v^{-2})$$

*so that for very $w \in W$, we have:*

$$\overline{T_w} = v^{-2\ell(w)} T_w + \text{ lower terms}$$

*dm,*

**Theorem 1.5** (Kazhdan-Lusztig)**.** *For any $w \in W$, there exists a unique element $C'_w \in \mathcal{H}_v(W)$ of the form:*

$$C'_w = v^{-\ell(w)}(T_w + \sum_{y<w} P_{y,w}(v^2)T_y)$$

*where $P_{y,w}(v^2) \in \mathbb{Z}[v^2]$ satisfying:*

1. $\deg P_{y,w} \leq (\ell(w) - \ell(y) - 1)/2$
2. $\overline{C'_w} = C'_w$

*Proof.* (of uniqueness) Suppose I have two choices: $C'_w, C''_w$, we must have:

$$C'_w - C''_w = v^{-\ell(w)} \sum_{y<w} Q_{y,w}(v^2)T_y$$

with $\deg Q_{y,w} \leq (\ell(w) - \ell(y) - 1)/2$. Choose $y_0$ be such that $\ell(y_0)$ maximal and $Q_{y_0,w} \neq 0$. We compute:

$$\overline{C'_w - C''_w} = v^{\ell(w)}(Q_{y_0,w}(v^{-2})\overline{T_{y_0}} + \text{ lower terms}) = v^{\ell(w)}(v^{-2\ell(y_0)}Q_{y_0,w}(v^{-2})\overline{T_{y_0}} + \text{lower terms})$$

Compare the coefficient in $T_{y_0}$, we see:

$$Q_{y_0,w}(v^2) = v^{2\ell(w)-2\ell(y_0)}Q_{y_0,w}(v^{-2}) = Q_{y_0,w}(v^2)$$

but LHS has dimension $< \ell(w)-\ell(y_0)$ with RHS has dimension $> \ell(w)-\ell(y_0)$, contradiction. □

It's easy to write down formula for some of $C'_w$:

1. $C'_1 = T_1$.
2. For $s \in S$, $C'_s = v^{-1}(T_s + 1)$. It satisfies $(C'_s)^2 = (v + v^{-1})C'_s$
3. For a finite Coxeter group $W$ with longest element $w_0$, we have:

$$C'_{w_0} = v^{-\ell(w_0)} \sum_{y \in W} T_y$$

$$(C'_{w_0})^2 = (v^{-\ell(w_0)} \sum_{y \in W} v^{2\ell(y)})C'_{w_0}$$

Note $\sum_{y \in W} v^{2\ell(y)}$ is the Poincare polynomial for the flag variety $\mathcal{B}$, if $W$ is a Weyl group.

# 2 Mixed Sheaves

The goal of this section is to study mixed sheaves on $\mathcal{B} \times \mathcal{B}$. We will drop the underline in $\underline{G}$ to still mean the algebraic group.

We will begin with some generality. Let $X$ be a scheme defined over finite field $k = \mathbb{F}_q$. Fix a prime $\ell$ coprime to $q$, and let $\mathcal{F}$ be an $\ell$-adic sheaf on $X$. The **geometric Frobenius** is defined to be:

$$\mathrm{Frob} = F_X \times \mathrm{id}_{\overline{k}} : X_{\overline{k}} \to X_{\overline{k}}$$

where $F_X$ is the $q$-absolute Frobenius on $X$ by raising everything to the $q$-th power. In particular if $x \in X(\mathbb{F}_q)$, then Frob fixes $x$. Now if $\mathcal{F}$ is an $\ell$-adic sheaf on $X$, and let $\overline{x} \in X(\overline{\mathbb{F}_q})$ corresponding to $x$, we have $\mathrm{Frob}^*$ acts on $\mathcal{F}_{\overline{x}}$.

**Example.** *A basic example of $\ell$-adic sheaf is Tate twist sheaf $\overline{\mathbb{Q}_\ell}(n)$. This will be a local system, so it suffices to specify a representation of $\pi_1^{\text{ét}}(X) \to \overline{\mathbb{Q}_\ell}^\times$. The Tate twists will be of arithmetic nature so that we require $\pi_1^{\text{ét}}(X_{\overline{k}})$ to act trivially. So it comes from a map:*

$$\mathrm{Gal}(\overline{\mathbb{F}_q}/\mathbb{F}_q) \to \overline{\mathbb{Q}_\ell}^\times$$

*But the absolute galois group is topologically generated by the arithmetic Frobenius $\sigma : t \mapsto t^q$, and we send this to $q^n$. More explicitly, this is the sheaf associated to $\ell^i$-th roots of unity on each finite level $i$.*

Now let's fix an embedding $\iota : \overline{\mathbb{Q}_\ell} \hookrightarrow \mathbb{C}$.

**Definition 2.1.** We call a sheaf $\mathcal{F} \in \mathrm{Shv}_c(X, \overline{\mathbb{Q}_\ell})$ **pointwise pure** of weight $w$ if for every closed point $x$ with residue field $\mathbb{F}_{q^n}$-point, we choose a geometric point $\overline{x}$ above it, and the eigenvalue of $\mathrm{Frob}_x = \mathrm{Frob}^n$, so that $\mathrm{Frob}_x$ will fix $x$ on $\mathcal{F}_{\overline{x}}$ has complex absolute value $q^{\frac{wn}{2}}$. A sheaf $\mathcal{F} \in \mathrm{Shv}_c(X, \overline{\mathbb{Q}_\ell})$ is called **mixed** if it has a finite filtration with each subquotients pure of some weights.

**Example.** *Tate twist $\overline{\mathbb{Q}_\ell}(n)$ is pure of weight $-2n$.*

**Definition 2.2.** A complex $\mathcal{F} \in D_c(X, \overline{\mathbb{Q}_\ell})$ is called **mixed** if each of its cohomology sheaf is mixed. We denote the full subcategories of $D_c(X, \overline{\mathbb{Q}_\ell})$ of such as:

$$D^{\mathrm{mix}}(X, \overline{\mathbb{Q}_\ell})$$

A complex $\mathcal{F} \in D^{\mathrm{mix}}(X, \overline{\mathbb{Q}_\ell})$ is called **pointwise $*$-pure** of weight $w$ if for each cohomology sheaf $\mathsf{H}^i(\mathcal{F})$, it is pure of weight $w + i$.

**Remark.** *A complex $\mathcal{F} \in D_G(\mathcal{B} \times \mathcal{B})$ is called mixed if for all $x \in (\mathcal{B} \times \mathcal{B})(\mathbb{F}_q)$, the action of $\mathrm{Frob}_x$ on the vector space $H^i(\mathcal{F}_{\overline{x}})$ has eigenvalues satisfying $|\bullet|_\iota = q^{n/2}$ for some $n \in \mathbb{Z}$.*

Note similar formulation hold for the $G$-equivariant version of the story. Now let's restrict our discussion to the setting $X = \mathcal{B} \times \mathcal{B}$.

**Definition 2.3.** A sheaf $\mathcal{F} \in D_G(\mathcal{B} \times \mathcal{B})$ is called **Tate** if the local systems $H^i(\mathcal{F}|_{\mathcal{O}(w)})$ is a successive extension of Tate twists $\overline{\mathbb{Q}_\ell}(n)$ for $n \in \mathbb{Z}$.

It is not hard to see:

**Proposition 2.4.** *If $\mathcal{F}$ is Tate and pure, then:*

1. *$H^i(\mathcal{F}) = 0$ if $i$ is odd.*
2. *The geometric Frobenius acts on $H^{2i}(\mathcal{F})$ by $q^i$ as eigenvalues.*

# 3 Hecke Category

We denote $\overline{G}, \overline{\mathcal{B}}$ be the base change of these objects to $\overline{\mathbb{F}_q}$. Let's start a particular instance of **sheaf-to-function correspondence**. For each $\mathcal{F} \in D_G(\mathcal{B} \times \mathcal{B})$, on can associate a function $f \in \mathrm{Fun}(\mathcal{B} \times \mathcal{B}(\mathbb{F}_q))^G$ via formula:

$$f(x,y) = \sum_{i \in \mathbb{Z}} (-1)^i \mathrm{Tr}(\mathrm{Frob}_{x,y}, H^i(\mathcal{F})_{\overline{(x,y)}})$$

This induces a map:

$$K_0(D_G(\mathcal{B} \times \mathcal{B})) \to \mathcal{H}_G$$

which is a ring homomorphism. In fact this can be extended to the following diagram:

$$\begin{array}{ccc}
K_0(D_{\bar{G}}(\bar{\mathcal{B}} \times \bar{\mathcal{B}})) & \xrightarrow{\chi} & \mathbb{Z}[W] \\
\uparrow & & \uparrow{\scriptstyle v \mapsto \pm 1} \\
K_0(D_G^{\mathrm{mix}}(\mathcal{B} \times \mathcal{B})) & \xrightarrow{\mathrm{WP}} & \mathcal{H}_v \\
\uparrow & & \\
K_0(D_G^{\mathrm{Tate}}(\mathcal{B} \times \mathcal{B})) & & \downarrow{\scriptstyle v \mapsto q^{1/2}} \\
\downarrow & & \\
K_0(D_G(\mathcal{B} \times \mathcal{B})) & \xrightarrow{\text{sheaf-function}} & \mathcal{H}_G
\end{array}$$

The reason for introducing $D_G^{\mathrm{Tate}}$ is that the natural map from the mixed complexes to all complexes will not make the diagram commute. The remaining goal of this section is to construct these maps and show the commutativity of the diagram.

The first maneuver is to introduce a monoidal structure on the sheaf categories, so that the $K$-groups will actually have a ring structure. Consider $\mathrm{pr}_{ij} : \mathcal{B} \times \mathcal{B} \times \mathcal{B} \to \mathcal{B} \times \mathcal{B}$ projection to the $(i,j)$-th coordinates. Given $\mathcal{F}_1, \mathcal{F}_2 \in D_G(\mathcal{B} \times \mathcal{B})$, we can thus consider the **convolution product**:

$$\mathcal{F}_1 * \mathcal{F}_2 \coloneqq \mathrm{pr}_{13,!}(\mathrm{pr}_{12}^* \mathcal{F}_1 \otimes \mathrm{pr}_{23}^* \mathcal{F}_2)$$

Then the unit is given by $\overline{\mathbb{Q}}_{\ell, \Delta\mathcal{B}}$. We can now construct the map $\chi$:

$$\chi : K_0(D_{\overline{G}}(\overline{\mathcal{B}} \times \overline{\mathcal{B}})) \to \mathbb{Z}[W]$$

$$\mathcal{F} \mapsto (w \mapsto \chi_{\text{euler}}(\mathcal{F}_{(B,B')}))$$

where $\chi_{\text{euler}}$ is simply the Euler character for a complex of vector spaces.

Note that $K_0(D_{\overline{G}}(\overline{B} \times \overline{B}))$ has a basis given by:

$$[i_{w,!}\overline{\mathbb{Q}_\ell}], \quad i_w : \mathcal{O}(w) \hookrightarrow \mathcal{B} \times \mathcal{B}$$

and by definition:

$$\chi([i_{w,!}\overline{\mathbb{Q}_\ell}]) = \mathbf{1}_w$$

This checks $\chi$ is an isomorphism of abelian groups.

**Proposition 3.1.** *The map $\chi$ is actually a ring isomorphism.*

*Proof.* It suffices to check:

1. $i_{w_1,!}\overline{\mathbb{Q}_\ell} * i_{w_2,!}\overline{\mathbb{Q}_\ell} \simeq i_{w_1w_2,!}\overline{\mathbb{Q}_\ell}$ when $\ell(w_1w_2) = \ell(w_1) + \ell(w_2)$
2. $[i_{s,!}\overline{\mathbb{Q}_\ell} * i_{s,!}\overline{\mathbb{Q}_\ell}] \simeq [i_{1,!}\overline{\mathbb{Q}_\ell}]$ for $s \in S$.

For 1, note that $\mathrm{pr}_{12}^* i_{w_1,!}\overline{\mathbb{Q}_\ell}$ is supported on $\mathcal{O}(w_1)\boxtimes\mathcal{B}$, $\mathrm{pr}_{23}^* i_{w_2,!}\overline{\mathbb{Q}_\ell}$ is supported on $\mathcal{B}\boxtimes\mathcal{O}(w_2)$. Note that if $\ell(w_1) + \ell(w_2) = \ell(w_1w_2)$ then:

$$p_{13} : \mathcal{O}(w_1) \boxtimes \mathcal{B} \cap \mathcal{B} \boxtimes \mathcal{O}(w_2) \to \mathcal{O}(w_1w_2)$$

will be an isomorphism. Indeed one see that:

$$\mathcal{O}(w_1) \boxtimes \mathcal{B} \cap \mathcal{B} \boxtimes \mathcal{O}(w_2) = \{(g_1B, g_2B, g_3B) \mid g_1^{-1}g_2 \in Bw_1B, g_2^{-1}g_3 \in Bw_2B\}$$

and note that $(Bw_1B) \cdot (Bw_2B) = B(w_1w_2)B$, so that the above $p_{13}$ is well-defined. This is an isomorphism because Bott-Samelson map is a birational isomorphism that is open on the open cell. This proves the first claim.

For the second part, let $p^{s,s} = p_{13}|_{\mathcal{O}(s)\times_{\mathcal{B}}\mathcal{O}(s)}$, then:

$$i_{s,!}\overline{\mathbb{Q}_\ell} * i_{s,!}\overline{\mathbb{Q}_\ell} = p_!^{s,s}\overline{\mathbb{Q}_\ell}$$

Again by a basic computation from the Bott-Samelson map, we see this is supported on $\mathcal{O}(s) \cup \mathcal{O}(1)$. Therefore, it suffices to compute the stalk at a point on $\mathcal{O}(s)$ and $\mathcal{O}(1)$. Now we have the diagram:

$$\begin{array}{ccc} \mathbb{G}_m & \hookrightarrow & \mathcal{O}(s) \times_{\mathcal{B}} \mathcal{O}(s) \\ \downarrow & & \downarrow \\ \{(B, B')\} & \hookrightarrow & \mathcal{O}(s) \cup \mathcal{O}(1) \end{array}$$

Proper base change tells me for $(B, B') \in \mathcal{O}(s)$:

$$(p_!^{s,s}\overline{\mathbb{Q}_\ell})|_{(B,B')} \simeq H_c^*(\mathbb{G}_m) = \begin{cases} \overline{\mathbb{Q}_\ell}, & \text{if } * = 1, 2; \\ 0, & \text{otherwise.} \end{cases}$$

so it vanishes in $K_0$. Similarly, we can compute:

$$(p_!^{s,s}\overline{\mathbb{Q}_\ell})|_{(B,B)} \simeq H_c^*(\mathbb{A}^1) = \begin{cases} \overline{\mathbb{Q}_\ell}, & \text{if } * = 2; \\ 0, & \text{otherwise.} \end{cases}$$

is one-dimensional. So that the result follows. □

We're now ready to construct WP. Before that we need the following fact for the above diagram makes sense:

1. Convolution product preserves mixedness.
2. Convolution product preserves Tate objects.

Recall the middle map:

$$K_0(D_G^{\mathrm{mix}}(\mathcal{B} \times \mathcal{B})) \xrightarrow{\mathrm{WP}} \mathcal{H}_v(W)$$
$$\mathcal{F} \to \sum_{w \in W} \varphi_w(v) T_w$$

Take $x \in \mathcal{O}(w)(\mathbb{F}_q)$, we have a grading on $H^i(\mathcal{F}_x)$:

$$H^i(\mathcal{F}_x)_j \coloneqq \bigoplus_{|\lambda| = q^{\frac{j}{2}}} H^i(\mathcal{F}_x)_\lambda$$

where $H^i(\mathcal{F}_x)_\lambda$is the generalized eigenspace of the Frobenius action on $H^i(\mathcal{F}_x)$ with eigenvalue $\lambda$. We then set:

$$\varphi_w(v) \coloneqq \sum_{i,j} (-1)^i \dim H^i(\mathcal{F}_x)_j v^j$$

# 4 Main Results of Kazhdan-Lusztig Theory

Choose a Borel $B_0 \subset G$. Using $B_0$ as a base point of $\mathcal{B}$, we have an equivalence of stacks:

$$G \backslash (\mathcal{B} \times \mathcal{B}) \xrightarrow{\sim} B_0 \backslash \mathcal{B}$$

by projection to the second factor. This induces an equivalence:

$$D_G(\mathcal{B} \times \mathcal{B}) \xrightarrow{\sim} D_{B_0}(\mathcal{B})$$

Recall we have the **Schubert cell**

$$\mathcal{B}_w = \{B \mid (B_0, B) \in \mathcal{O}(w)\} = B_0 \backslash B_0 w B_0 \subset \mathcal{B}$$

and **Schubert variety**: $\overline{\mathcal{B}_w}$. We set $\mathscr{C}_w = i_{w,!*}(\overline{\mathbb{Q}_\ell})$, so that $\mathscr{C}_w|_{\mathcal{B}_w} \simeq \overline{\mathbb{Q}_\ell}$. This is related to standard IC via:

$$\mathrm{IC}_w = \mathscr{C}_w[\ell(w)](\frac{\ell(w)}{2})$$

**Theorem 4.1** (Kazhdan-Lusztig)**.** *For all $w \in W$,*

$$\mathrm{WP}(\mathscr{C}_w) = \sum_{y \leq w} P_{y,w}(v^2) T_y = v^{\ell(w)} C'_w$$

**Theorem 4.2.** *For all $w \in W$, $\mathscr{C}_w$ is pointwise $*$-pure and Tate. In particular $\mathscr{C}_w$ also has parity vanishing. The sheaf-to-function correspondence sends $\mathscr{C}_w$ to*

$$\sum_{y \leq w} P_{y,w}(q) T_y \in \mathcal{H}_G$$

**Proposition 4.3.** *Any mixed sheaf $\mathcal{F} \in D^{\mathrm{mix}}_{B_0}(\mathcal{B})$ is a successive extension of*

$$i_{w!}\overline{\mathbb{Q}}_\ell \otimes V_w$$

*where $V_w$ is the Frobenius module*

It is clear from the construction that:

$$\mathrm{WP}([i_{w!}\overline{\mathbb{Q}_\ell}]) = T_w$$

Therefore, the above proposition implies that for any $\mathcal{F} \in D^{\mathrm{mix}}_{B_0}(\mathcal{B})$, in the Grothendieck group:

$$[\mathcal{F}] = \sum_{w \in W} [\mathcal{F}|_{x_w}] \cdot [i_{w!}\overline{\mathbb{Q}_\ell}]$$

It follows that:

$$\mathrm{WP}(\mathcal{F}) = \sum_{w \in W} \mathrm{WP}(\mathcal{F}|_{x_w}) \cdot T_w$$

To make sense of this, we view:

$$\mathcal{F}|_{x_w} \in D^{\mathrm{mix}}(\mathrm{Spec}\ \mathbb{F}_q)$$

and therefore:

$$\mathrm{WP}(\mathcal{F}|_{x_w}) \in \mathbb{Z}[v^{\pm 1}]$$

Now plug in $\mathcal{F} = \mathscr{C}_w$, we see that:

$$P_{y,w}(v^2) = \mathrm{WP}(\mathscr{C}_w|_{x_y}) = \sum_{j \geq 0} \dim H^{2j}(\mathscr{C}_w|_{x_y}) v^j$$

By the purity and parity of $\mathscr{C}_w$. Therefore for computing the Kazhdan Lusztig polynomial, we actually don't care about the Tate twist.

# 5 Exampe: $\mathbf{SL}_4$

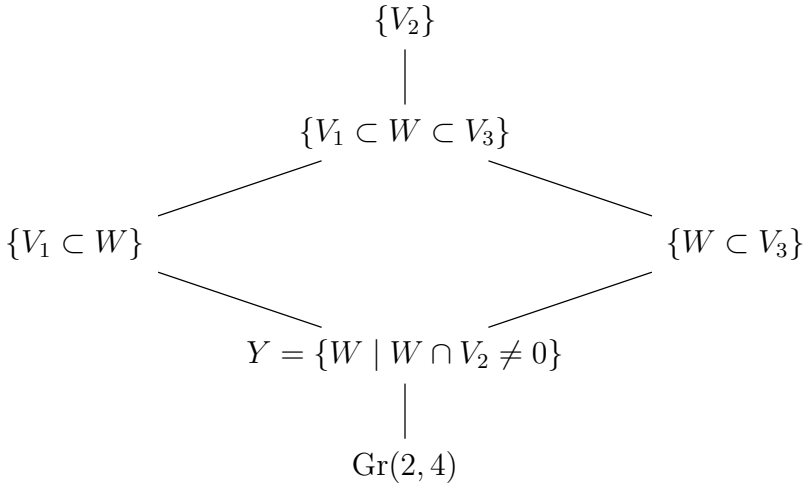
We will compute the first nontrivial Kazhdan Lusztig polynomial. We consider $G = \mathrm{SL}_4$ acting on the partial flag variety $\mathrm{Gr}(2,4) \simeq G/P$, for:

$$P = \begin{pmatrix} * & * & * & * \\ * & * & * & * \\ & & * & * \\ & & * & * \end{pmatrix}$$

Let's fix a standard flag:

$$V_1 \subset V_2 \subset V_3 \subset V_4 = V$$

with respect to ordered basis $\{e_1, e_2, e_3, e_4\}$. We pick the base point to be $V_2$ and it has stablizer $P$, this gives the above realization $\mathrm{Gr}(2,4) \simeq G/P$. In this instance we have six Schubert varieties:




Now consider an open neighborhood of $V_2$:

$$U \simeq \mathbb{A}^4 \hookrightarrow \mathrm{Gr}\ (2,4)$$

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mapsto \begin{pmatrix} 1 & & & \\ & 1 & & \\ a & b & 1 & \\ c & d & & 1 \end{pmatrix} \cdot V_2$$

Note that $M = U \cap Y \iff \det M = 0$. So that $Y$ has a singularity at origin. Now to compute $\mathrm{IC}(Y)$, we must resolve singularity. Consider:

$$\tilde{Y} = \{(L, W) \in \mathbb{P}(V) \times \mathrm{Gr}(2,4) \mid L \subset W \cap V_2 \neq 0\}$$

Then one sees that

$$\pi : \tilde{Y} \to Y$$

is a resolution of singularity: it is isomorphism outside at $V_2$, and the fibre at $V_2$ is $\mathbb{P}(V_2)$. This actually shows $\pi$ is a small map, therefore we know:

$$R\pi_*\underline{\mathbb{Q}_\ell}[3] = \mathrm{IC}(Y)$$

Again by proper base change, we see (ignoring the Tate twist):

$$(R\pi_*\underline{\mathbb{Q}_\ell})_W = \begin{cases} \mathbb{Q}_\ell, & \text{if } W \neq V_2; \\ H^*(\mathbb{P}^1) = \mathbb{Q}_\ell \oplus \mathbb{Q}_\ell[-2], & \text{if } W = V_2. \end{cases}$$

Since our previous discussion is on the full flag variety, we need the following diagram:

$$\begin{array}{ccccc} \mathcal{B} = \mathcal{F}\ell_4 & \hookleftarrow & \overline{\mathcal{B}_w} & \hookleftarrow & \overline{\mathcal{B}_y} = \mathbb{P}^1 \times \mathbb{P}^1 \\ \downarrow{\scriptstyle \pi} & & \downarrow & & \downarrow \\ \mathrm{Gr}(2,4) & \hookleftarrow & Y & \hookleftarrow & \{W = V_2\} \end{array}$$

Now choose $x_y \in \mathcal{B}_y$. Now we have:

$$\pi^*\mathrm{IC}(Y)[2] = \mathrm{IC}_w \implies \mathscr{C}_w = \pi^*\mathrm{IC}(Y)[-3]$$

Therefore:

$$\mathscr{C}_w|_{x_y} = \mathrm{IC}(Y)[-3]|_{V_2} = \mathbb{Q}_\ell \oplus \mathbb{Q}_\ell[-2]$$

This tells us the Kazhdan Lusztig polynomial:

$$P_{y,w}(v^2) = 1 + v^2$$

# 6 Bott-Samelson Resolution

Let $w \in W$ be a Weyl group element and fix a reduced expression

$$w = s_1 \cdots s_\ell$$

We also fix a Borel $B_0$. We define:

$$\mathrm{BS}(s_1, \cdots, s_\ell) = \{B_0 \xrightarrow{\leq s_1} B_1 \xrightarrow{\leq s_2} \cdots \xrightarrow{\leq s_\ell} B_\ell\} \subset \mathcal{B}^{\ell+1}$$

where $\leq s_i$ means the relative position of the two Borels are either 1 or $s_i$. Then we have a natural map:

$$\mathrm{BS}(s_1, \cdots, s_\ell) \to \overline{\mathcal{B}_w}$$
$$(B_0, \cdots, B_\ell) \mapsto B_\ell$$

This is a resolution of singularity. We can generalize this construction to partial flag varieties as following. Given $G/B \to G/P$, we can form the Weyl group $W_P$ corresponding to parabolic subgroup $P$. It is known the following diagram commutes:

$$\begin{array}{ccc} B\backslash G/B & \twoheadrightarrow & B\backslash G/P \\ \simeq \uparrow & & \uparrow \simeq \\ W & \twoheadrightarrow & W/W_P \end{array}$$

Now given each element in $x \in W/W_P$, it will have several lifts. There will be a maximal one $w$ and minimal one $v$ related by:

$$w = vw_{0,P}$$

where $w_{0,P}$ is the longest element in $W_P$. Then the following diagram is Cartesian for $w$ the maximal lift of $[w]$:

$$\begin{array}{ccc} G/B & \hookleftarrow & \overline{\mathcal{B}_w} \\ \downarrow & & \downarrow \\ G/P & \hookleftarrow & \overline{\mathcal{B}^P_{[w]}} \end{array}$$

A Bott-Samelson resolution for $\mathcal{B}^P_{[w]}$ is equivalent to picking a reduced word expression for the minimal lift $v$ of $[w]$:

$$v = s_1 \cdots s_m$$

$$\mathrm{BS}(s_1, \cdots, s_m) = P_{s_1} \times_B P_{s_2} \times_B \cdots P_{s_m}/B$$

Now by picking a reduced expression for $w_{0,P} = t_1 \cdots t_n$, we get a Bott-Samelson resolution for $\overline{\mathcal{B}_w}$. Everything fits into:

$$\begin{array}{ccccc} \mathrm{BS}(t_1, \cdots, t_n) & \longrightarrow & \mathrm{BS}(s_1, \cdots, s_m, t_1, \cdots, t_n) & \longrightarrow & \mathrm{BS}(s_1, \cdots, s_m) \\ \downarrow & & \downarrow & & \downarrow \\ P/B & \longrightarrow & \overline{\mathcal{B}_w} & \longrightarrow & \overline{\mathcal{B}^P_w} \end{array}$$

with each column a BS resolution, while each row is a fibre bundle.

**Example.** *Let's explicitly compute a Bott-Samelson resolution of* $\overline{\mathcal{B}_w} = \pi^{-1}(Y)$ *in our previous example. According to the above discussion, we know* $w$ *is the maximal lift for the coset of* $W/W_P$ *corresponding ot* $Y$*. It's not hard to see:*

$$w = s_1 s_3 s_2 s_3 s_1$$

*Now consider:*

$$\nu : \mathrm{BS}(s_1, s_3, s_2, s_3, s_1) \to \overline{\mathcal{B}_w}$$

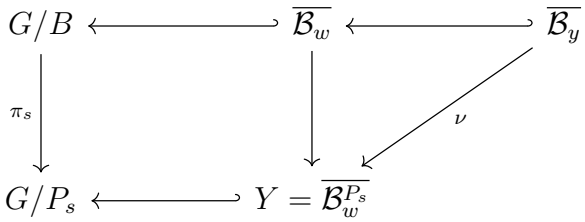
*and pick* $x = (F_1' \subset V_2 \subset F_3') \in \overline{\mathcal{B}_w}$*, we see:* $\nu^{-1}(x)$ *is parametrizing* $(F_1, F_3)$ *such that* $F_1 \subset V_2, F_3 \supset V_2$*. Therefore:*

$$\nu^{-1}(x) \simeq \mathbb{P}^1 \times \mathbb{P}^1$$

# 7 Proof of Main Results

Let's start from:

*Proof.* (of purity and parity) We induct on $\ell(w)$. Suppose $w = ys$ for simple reflection $s$ and $\ell(w) > \ell(y)$. We look at:




where the left square is pullback and $\nu$ is birational. Let $\mathscr{C}_Y = \mathrm{IC}(Y)[-\dim Y]$. Since $\pi_s$ is a smooth $\mathbb{P}^1$-fibration:

$$\pi_s^* \mathscr{C}_Y = \mathscr{C}_w$$

So to check purity and parity of $\mathscr{C}_w$, it suffices to check for $\mathscr{C}_Y$. We wish to check purity and parity for each $\mathscr{C}_Y|_x$, $x \in Y$.

Note for each $x \in Y$, the fibre $\nu^{-1}(x)$ is either a point or $\mathbb{P}^1$, stratified as $\mathbb{A}^1 \cup \{\infty\}$. If $\nu^{-1}(x)$ is a point $\widetilde{x}$, then proper base change tells:

$$\mathscr{C}_y|_{\widetilde{x}} = (R\nu_* \mathscr{C}_y)|_x$$

But decomposition theorem tells me $\mathscr{C}_Y$ is a direct summand inside $R\nu_* \mathscr{C}_y \implies \mathscr{C}_Y|_x$ is a direct summand of $\mathscr{C}_y|_{\widetilde{x}}$, so the parity and purity follows by induction hypothesis.

Suppose on the other hand that $\nu^{-1}(x) = \mathbb{P}^1$. Then again by proper base change and decomposition theorem, we know $\mathscr{C}_Y|_x$ is a direct summand of

$$H^*(\mathbb{P}^1, \mathscr{C}_y|_{\nu^{-1}(x)}) = (R\nu_* \mathscr{C}_y)|_x$$

Therefore it remains to check purity and parity of $H^*(\mathbb{P}^1, \mathscr{C}_y|_{\nu^{-1}(x)})$. Now we have excision triangle:

$$H_c^*(\mathbb{A}^1, \mathscr{C}_y|_{\mathbb{A}^1}) \to H^*(\mathbb{P}^1, \mathscr{C}_y|_{\nu^{-1}(x)}) \to H^*(\{\infty\}, \mathscr{C}_y|_\infty) \to$$

The third term satisifies purity and parity from the purity and parity on $\mathscr{C}_y$. But $\mathscr{C}_y|_{\mathbb{A}^1}$ is geometrically a constant sheaf, we have:

$$H_c^*(\mathbb{A}^1, \mathscr{C}_y|_{\mathbb{A}^1}) \simeq H_c^*(\mathbb{A}^1) \otimes (\mathscr{C}_y|_{\mathbb{A}^1}) \simeq \underline{\mathbb{Q}_\ell}[-2](-1) \otimes (\mathscr{C}_y|_{\mathbb{A}^1})$$

again satisifes purity and parity. Therefore we conclude the middle term also satisifes purity and parity. □

Next we need to prove our claimed formula for the map WP. The key input is that the Verdier duality functor dualizes the bar involution.

**Proposition 7.1.** *For all $\mathcal{F} \in D^{\mathrm{mix}}_{B_0}(\mathcal{B})$, we have:*

$$\mathrm{WP}(\mathbb{D}\mathcal{F}) = \overline{\mathrm{WP}(\mathcal{F})}$$

*Proof.* Let $s$ be a simple reflection. We first prove the the result for $\mathcal{F} = i_{s!}\overline{\mathbb{Q}_\ell}$. Namely:

$$\mathrm{WP}(\mathbb{D}(i_{s!}\overline{\mathbb{Q}_\ell})) = \overline{\mathrm{WP}(i_{s!}\overline{\mathbb{Q}_\ell})}$$
$$\iff \mathrm{WP}(i_{s*}\overline{\mathbb{Q}_\ell}[2](1)) = \overline{T_s} = v^{-2}T_s + (v^{-2}-1)T_1$$

For the LHS, pick a point $x \in \mathcal{O}(s)$, we have:

$$(i_{s*}\overline{\mathbb{Q}_\ell}[2](1))_x = \overline{\mathbb{Q}_\ell}[2](1)$$

concentrated in degree $-2$ with Frobenius eigenvalue $q^{-2}$, this gives the term $v^{-2}T_s$. Pick the point $y \in \mathcal{O}(1)$:

$$H^*((i_{s*}\overline{\mathbb{Q}_\ell}[2](1))_y) = H^*(\mathbb{G}_{m,\overline{\mathbb{F}_q}}, \overline{\mathbb{Q}_\ell}[2](1)) = \begin{cases} \overline{\mathbb{Q}_\ell}(1), & \text{if } i = -2; \\ \overline{\mathbb{Q}_\ell}, & \text{if } i = -1. \end{cases}$$

Thus this gives the term $(v^{-2}-1)T_1$. This proves the claim in this case. Now the general case follows from WP and $\mathbb{D}$ are both compatible with convolution. □

*Proof.* (of Kazhdan-Lusztig). We must show:

1. $v^{-\ell(w)}\mathrm{WP}(\mathscr{C}_w)$ is bar invariant.
2. $\mathrm{WP}(\mathscr{C}_w) = T_w + \sum_{y<w} P_{y,w}(v^2)T_y$ for $\deg P_{y,w} \leq (\ell(w) - \ell(y) - 1)/2$.

THe first condition is easy to check. Recall:

$$\mathscr{C}_w = \mathrm{IC}_w[-\ell(w)](-\frac{\ell(w)}{2})$$

and $\mathbb{D}(\mathrm{IC}_w) = \mathrm{IC}_w$

$$\overline{\mathrm{WP}(\mathscr{C}_w)} = \mathrm{WP}(\mathbb{D}(\mathrm{IC}_w[-\ell(w)](-\frac{\ell(w)}{2}))) = \mathrm{WP}(\mathrm{IC}_w[\ell(w)](\frac{\ell(w)}{2})) = \mathrm{WP}(\mathscr{C}_w[2\ell(w)](\ell(w)))$$

It follows:

$$\overline{\mathrm{WP}(\mathscr{C}_w)} = v^{-2\ell(w)}\mathrm{WP}(\mathscr{C}_w) \implies \overline{v^{-\ell(w)}\mathrm{WP}(\mathscr{C}_w)} = v^{-\ell(w)}\mathrm{WP}(\mathscr{C}_w)$$

as required. For the second part, the basic property of IC sheaf tells me $\mathrm{IC}_w|_{\mathcal{O}(y)}$ is supported in degree $\leq -\ell(y)-1$. Therefore $\mathscr{C}_w$ is supported in degrees $\leq \ell(w)-\ell(y)-1$. The definition of WP makes sure the coefficient before $T_y$ for $y < w$ is:

$$\sum_{0 \leq i \leq \ell(w)-\ell(y)-1} (-1)^i \dim H^i(\mathscr{C}_w|_y)_j v^j$$

But by **purity and parity**, we know the necessary condition for $H^i(\mathscr{C}_w|_y)_j$ to be nonzero is $i = j$ is even. This shows the above polynomial as degree $\leq \ell(w) - \ell(y) - 1$ and only has even terms. This concludes the proof. □

**Remark.** *In fact the above proof give the* **formula** *for Kazhdan-Lusztig polynomial:*

$$P_{y,w}(v^2) = \sum_{i \geq 0} \dim H^i(\mathscr{C}_w|_y) v^i$$

# 8 Multiplication Table

**Definition 8.1.** Let $s, w \in W$ with $s$ simple reflection. We call $w$ **left $s$-vertical** if $\ell(sw) < \ell(w)$; otherwise call it **left $s$-horizontal**.

**Theorem 8.2.** *Let $s, w \in W$ with $s$ a simple reflection. Then:*

$$C'_s C'_w = \begin{cases} (v + v^{-1}) C'_w, & \text{if } \ell(sw) < \ell(w); \\ \sum_{y \leq sw, \ell(sy) < \ell(y)} \mu(y,w) C'_y, & \text{otherwise.} \end{cases}$$

*where $\mu(y,w)$ is certain nonnegative integer. In fact $\mu(y,w)$ equals the coefficient of $q^{\frac{1}{2}(\ell(w)-\ell(y)-1)}$ in $P_{y,w}(q)$.*

To proceed, Let's fix a Borel $B_0$. We need to understand the convolution product in the realization of derived category of sheaves on the quotient stack: $D(B_0 \backslash G / B_0)$. We need to start with the **convolution diagram**:

$$B_0 \backslash G / B_0 \times B_0 \backslash G / B_0 \xleftarrow{p} B_0 \backslash G \times^{B_0} G / B_0 \xrightarrow{m} B_0 \backslash G / B_0$$

For each sheaf operation we mean its derived version. Now given $\mathcal{F}, \mathcal{G} \in D(B_0 \backslash G / B_0)$, we set:

$$\mathcal{F} * \mathcal{G} = m_* p^* (\mathcal{F} \boxtimes \mathcal{G})$$

It can be shown this construction gives the same convolution product as our previous construction.

Now we can compute:

$$\mathscr{C}_s * \mathcal{F}$$

by looking at the following convolution diagram:

$$\begin{array}{ccccc} B\backslash P_s/B \times B\backslash G/B & \xleftarrow{p} & B\backslash P_s \times^B G/B & \xrightarrow{m} & B\backslash G/B \\ \downarrow{\scriptstyle \mathrm{pr}'_2} & & \downarrow{\scriptstyle \mathrm{pr}_2} & & \downarrow{\scriptstyle \pi_s} \\ B\backslash G/B & \xleftarrow{=} & B\backslash G/B & \xrightarrow{\pi_s} & P_s\backslash G/B \end{array}$$

Note that:

$$\mathscr{C}_s = \overline{\mathbb{Q}_\ell}|_{B\backslash P_s/B}$$

Hence:

$$\mathscr{C}_s \boxtimes \mathcal{F} = \mathrm{pr}_2'^* \mathcal{F}$$

Therefore:

$$\mathscr{C}_s * \mathcal{F} = m^* p^* \mathrm{pr}_2'^* \mathcal{F} = m_* \mathrm{pr}_2^* \mathcal{F} = \pi_s^* \pi_{s*} \mathcal{F}$$

where the last step is proper base change.

Now we want to specialize $\mathcal{F} = \mathscr{C}_w$. If $\ell(sw) < \ell(w)$, we know that:

$$\mathscr{C}_w = \pi_s^* \mathscr{C}_{[w]}$$

for $\mathscr{C}_{[w]}$ the sheaf supported on the Schubert variety corresponding to $[w]$ on $P_s \backslash G$. Therefore projection formula tells me:

$$\pi_{s*} \pi_s^* \mathscr{C}_{[w]} = \mathscr{C}_{[w]} \otimes \pi_{s*} \overline{\mathbb{Q}_\ell} = \mathscr{C}_{[w]} \oplus \mathscr{C}_{[w]}[-2](-1)$$

since $\pi_s$ is a smooth $\mathbb{P}_1$-fibration. It follows that:

$$\mathscr{C}_s * \mathscr{C}_w = \mathscr{C}_w \oplus \mathscr{C}_w[-2](-1)$$

Apply WP we get the desired result.

For the second case, we must check:

$$\pi_{s*} \mathrm{IC}_w = \mathrm{IC}_{[w]} \oplus \bigoplus_{y < sw; \ell(sy) < \ell(y)} \mathrm{IC}_{[y]}^{\oplus \mu(y,w)}$$

Once we prove this, then:

$$\mathrm{IC}_s * \mathrm{IC}_w = \pi_s^* (\mathrm{IC}_{[w]} \oplus \bigoplus_{y < sw; \ell(sy) < \ell(y)} \mathrm{IC}_{[y]}^{\oplus \mu(y,w)})[1](\frac{1}{2}) = \mathrm{IC}_{sw} \oplus \bigoplus_{y < sw; \ell(sy) < \ell(y)} \mathrm{IC}_y^{\oplus \mu(y,w)}$$

Apply WP:

$$(-1) C_s' \cdot (-1)^{\ell(w)} C_w' = (-1)^{\ell(sw)} C_{sw}' + \sum_{y < sw; \ell(sy) < \ell(y)} (-1)^{\ell(y)} \mu(y,w) C_y'$$

By the parity and our explicitly description of $\mu(y,w)$, we will see $\mu(y,w) = 0$ for $\ell(w) - \ell(y)$ is even. Therefore the sign issue goes away and we get our desired result.

Therefore it remains to check our formula for $\pi_{s*} \mathrm{IC}_w$. This can be done by analyzing the the degrees of support of

$$\pi_{s*} \mathrm{IC}_w |_{[y]}$$

and the upshot is that we can check that $\pi_{s*} \mathrm{IC}_w$ is a perverse sheaf (no shift). Then the result follows from Decomposition theorem.

# 9 Modules for Hecke Algebra

The easiest module for Hecke algebra $\mathcal{H}_v(W)$ is the left regular representation. To understand this representation is the same as understanding the multiplication table as we did in the previous section. We will package these informtion into a combinatorial structure known as a $W$-graph.

**Definition 9.1.** The $W$**-graph** attached to the left $\mathcal{H}_v(W)$-module $\mathcal{H}_v(W)$ consists of the following data:

1. A vertex set indexed by $W$: think of $w \in W$ corresponding to $C'_w$.
2. A labeling of each vertex $w$ by the left descent set:
$$\mathcal{D}(w) \coloneqq \{s \in S : \ell(sw) < \ell(w)\}$$
3. A set of (undirected) edges, where $y-w$ is an edge if $y < w$, $P_{y,w}$ has maximal possible degree $\frac{1}{2}(\ell(w) - \ell(y) - 1)$ and $\mathcal{D}(y) \not\subset \mathcal{D}(w)$.
4. A labelling of the edge $y-x$ be the nonnegative integer $\mu(y, w)$.

In general, suppose we're given a data:
$$(V, E, \{\mathcal{D}(y)\}_{y\in V}, \{\mu(y, w)\}_{(y-w)\in E})$$
where:

1. $V$ is a vertex set.
2. $E$ is an (unordered) edge set.
3. Each $\mathcal{D}(y) \subset S$.
4. $\mu(y, w)$ assigns to each ordered pair $(y, w)$ with $y-w \in E$ an integer $\mu(y, w)$.

One can hope to give $M = \mathbb{Z}[v, v^{-1}][V]$ (a free $\mathbb{Z}[v, v^{-1}]$-module with basis $\{C_y\}_{y\in V}$) a $\mathcal{H}_v(W)$-module structure by:
$$C'_s \cdot C_y = \begin{cases} (v + v^{-1})C_y, & \text{if } s \in \mathcal{D}(y); \\ \displaystyle\sum_{x-y, s\in\mathcal{D}(x)} \mu(x, y)C_x & \text{otherwise.} \end{cases}$$
We will call this data a $W$**-graph** if this indeed defines a $\mathcal{H}_v(W)$-action on $M$.

**Definition 9.2.** We call a $G$-variety $X$ **sphericial** if it has finitely many $B_0$-orbits.

Given a spherical variety $X$, we look at the following convolution diagram:
$$B_0 \setminus G/B_0 \times B_0 \setminus X \xleftarrow{\quad p \quad} B_0 \setminus G \times^{B_0} X \xrightarrow{\quad a \quad} B_0 \setminus X$$
This defines a convolution action:
$$D(B_0\backslash G/B_0) \times D(B_0 \setminus X) \to D(B_0 \setminus X)$$
$$(\mathcal{F}, \mathcal{M}) \mapsto \mathcal{F} * \mathcal{M} = a_* p^*(\mathcal{F} \otimes \mathcal{M})$$

Assume for simplicity each geometric $B_0$-orbit on $X$ contains a $\mathbb{F}_q$-point (hence the orbit is defined over $\mathbb{F}_q$).

Let $M_X$ be the free $\mathbb{Z}[v, v^{-1}]$-module with basis $T_{\mathcal{O},\mathcal{L}}$ indexed by a geometric $B_0$-orbit $\mathcal{O}$ and an irreducible $B_0$-equivariant $\overline{\mathbb{Q}_\ell}$-local system on $\mathcal{O}$, up to twisting the Frobenius structure by scalars. We assume that each $\mathcal{L}$ remains irreducible over $\mathcal{O}_{\overline{\mathbb{F}_q}}$.

As before, we will have a weight polynomial map:

$$\mathrm{WP} : K_0(D^{\mathrm{mix}}_{B_0}(X)) \to M_X$$

This will make $M_X$ a $\mathcal{H}_v(W)$-module via the convolution we constructed before. In this setup, $M_X$ has two type of basis:

1. Standard basis: $T_{\mathcal{O},\mathcal{L}} = \mathrm{WP}(i_{\mathcal{O}!}\mathcal{L})$.
2. Kazhdan-Lusztig basis: $\mathrm{WP}(\mathrm{IC}_{\mathcal{O},\mathcal{L}})$

In this setup, part of the data of the $W$-graph for $M_X$ will be:

- $V = \{(\mathcal{O}, \mathcal{L})\}$
- $\mathcal{D}(\mathcal{O}, \mathcal{L})$ consists of those $s$ such that: $\mathcal{O}$ is $s$-vertical ($\dim(P_s\mathcal{O}) = \dim \mathcal{O}$), and $\mathcal{L}$ extends to a $P_s$-equivariant local system on $P_s\mathcal{O}$.

To study edges and $\mu$, we need to carefully study the convolution product. Let

$$\mathrm{act}_s : P_s \times^B X \to X$$

then

$$\mathrm{IC}_s * \mathrm{IC}_{\mathcal{O},\mathcal{L}} = \mathrm{act}_{s*}(\widetilde{\mathrm{IC}_{\mathcal{O},\mathcal{L}}}[1](\frac{1}{2}))$$

where $\widetilde{\mathrm{IC}_{\mathcal{O},\mathcal{L}}} = \mathrm{IC}(P_s \times^B \overline{\mathcal{O}}, \mathcal{L})$. As a consequence of decomposition theorem, we know this should be a direct sum of shifts of simple perverse sheaves.

**Fact**: If $(\mathcal{O}, \mathcal{L})$ is $s$-vertical, then:

$$\mathrm{IC}_s * \mathrm{IC}_{\mathcal{O},\mathcal{L}} \simeq \mathrm{IC}_{\mathcal{O},\mathcal{L}}[1](\frac{1}{2}) \oplus \mathrm{IC}_{\mathcal{O},\mathcal{L}}[-1](-\frac{1}{2})$$

**Fact**: If $(\mathcal{O}, \mathcal{L})$ is $s$-horizontal, then:

$$\mathrm{IC}_s * \mathrm{IC}_{\mathcal{O},\mathcal{L}} \simeq \bigoplus_{\mathcal{O}',\mathcal{L}'} \mathrm{IC}_{\mathcal{O}',\mathcal{L}'}^{\oplus \mu((\mathcal{O}',\mathcal{L}'),(\mathcal{O},\mathcal{L}))}$$

where the sum runs over pairs $(\mathcal{O}', \mathcal{L}')$ such that $s \in \mathcal{D}(\mathcal{O}', \mathcal{L}')$ and $\mathcal{O}' \subset \overline{P_s\mathcal{O}}$ subject to additional constraint:

- $\mathrm{Hom}(\mathcal{L}', \mathrm{IC}_{\mathcal{O},\mathcal{L}}|_{\mathcal{O}'})$ has nonzero cohomology in degree $-\ell(\mathcal{O}') - 1$ if $\mathcal{O}' \subset \overline{\mathcal{O}}$ (this is analogous to the condition that $P_{y,w}$ attains is maximal degree)
- and there is an additional condition depending on $\mathcal{L}$ if $\mathcal{O} \neq \mathcal{O}' \subset P_s\mathcal{O}$

**Remark.** *To prove these facts, we need to understand the $B_0$-orbits on $P_s\mathcal{O}$. Now if $\mathcal{O} = B/H$, then we need to understand $B_0 \setminus P_s/H \simeq \mathbb{P}^1/H$. We know this must be finite since $X$ is spherical variety. In this setting we have the following classification:*

1. *Type $U$. $\mathbb{P}^1/H \simeq \mathbb{P}^1/U$ has two orbits: $\{0\}, \mathbb{P}^1 \setminus \{0\}$.*
2. *Type $G$: one orbit $\mathbb{P}^1$.*
3. *Type $T$: three orbits $\{0\}, \{\infty\}, \mathbb{P}^1 \setminus \{0, \infty\}$.*
4. *Type $N = N_G(T)$: two orbits $\{0, \infty\}, \mathbb{P}^1 \setminus \{0, \infty\}$.*

# 10 Cells

We return to the $W$-graph of $\mathcal{H}_v(W)$ as a left module over itself. We orient the edges of the $W$-graph by declaring $w \to y$ if $C'_y$ appears in $C'_s C'_w$ for some $s \in \mathcal{D}(y) \setminus \mathcal{D}(w)$. This gives a preorder $\succcurlyeq_L$ on $W$ generated by $w \succcurlyeq y$ if $w \to y$. This preorder partitions $W$ into equivalence classes, where $x \sim_L y$ if $x \succcurlyeq_L y$ and $y \succcurlyeq_L x$. These equivalence classes are called **left cells** of $W$.

We can define the left cells of $W$ algebraically. The left ideals of $\mathcal{H}_v(W)$ spanned by a subset $\{C'_w\}_{w \in W}$ form a poset by inclusion. If $I' \subset I$ is a minimal inclusion, then as a left $\mathcal{H}_v(W)$-module $I/I'$ is spanned by $C'_w$ for $w$ lying in a single left cell. Conversely, given every left cell $X \subset W$, we can build a left $\mathcal{H}_v(W)$-module as follows. As a $\mathbb{Z}[v, v^{-1}]$-module it has basis $\{C'_x \mid x \in X\}$. Multiplication by $C'_s \in \mathcal{H}_v(W)$ is given by the standard multiplication mudulo:

$$\{C'_y : y \prec_L X\}$$

**Theorem 10.1** (Kazhdan-Lusztig)**.** *For $W \in S_n$, each left cell module is a simple module (after applying $- \otimes_{\mathbb{Z}[v,v^{-1}]} \overline{\mathbb{Q}(v)}$)*

In general, the notion of left cells makes sense for any spherical variety $X$. Similarly, one can define the notion of **right cells** or **two-sided cells** of $W$. In particular, the two-sided cell corresponds to the $G \times G$-spherical variety $G \times G/\Delta G \simeq G$. The cells are again given by $I/I'$ for $I' \subset I$ minimal inclusion of two-sided ideals.

**Example.** *Recall the* **Robinson-Schensted correspondence***, which gives a bijection:*

$$W = S_n \longleftrightarrow \{(\lambda, P, Q) : P, Q \text{ SYT of shape } \lambda \vdash n\}$$

*In terms of this correspondence:*

1. *The left cells of $S_n$ are $\sigma \in S_n$ with the same $Q$;*
2. *The right cells of $S_n$ are $\sigma \in S_n$ with the same $P$;*
3. *The two-sided cells of $S_n$ are those with the same $\lambda$*

*Given a SYT $Q$ of shape $\lambda$, let $X_Q$ be the corresponding left cell of $S_n$. View left cell module corresponding to $X_Q$ as a left $\mathbb{Z}[S_n]$-module by setting $v = 1$. It is known that this gives the irreducible $S_n$-module $V_\lambda$ corresponding to $\lambda$, and the analogous construction for right cells yields the dual $V_\lambda^*$. This gives an $\mathcal{H}_v(W)$-version of Peter-Weyl:*

$$\mathbb{C}[S_n] = \oplus_{\lambda \vdash n} V_\lambda \otimes V_\lambda^*$$

# 11 A Big Example

Let $G = \mathrm{GL}_n(\mathbb{C})$, $W = S_n$. In this section, we will focus on computing explicit data respect to the spherical variety $X = G/B \times \mathbb{C}^n = \mathcal{F}\ell_n \times \mathbb{C}^n$.

## 11.1 Parametrization of Orbits

Let's fix a base point $wB/B \in G/B$. The $B$-action has stablizer:

$$B_w = B \cap wBw^{-1}$$

So we need to classify $B_w$-orbits on $\mathbb{C}^n$. Note that explicitly:

$$B_w = T \cdot \prod_{i<j, w^{-1}(i)<w^{-1}(j)} U_{ij}$$

Therefore the $B_w$-orbits on $\mathbb{C}^n$ are parametrized by subset $S_w \subset \{1, 2, \cdots, n\}$ such that

$$i < j \in S_w \implies w^{-1}(i) > w^{-1}(j)$$

Hence $B$ orbits on $X$ are parametrized by:

$$(w, S_w) \in W \times \mathcal{P}([n])$$

with $S_w$ satisfying the above condition.

For the future computation for the IC sheaves, we also need to understand the stablizer of each orbit. Let's pick an element

$$(wB/B, e_{i_1} + \cdots e_{i_k}) \in \mathcal{O}_{(w,S_w)}$$

where $S_w = \{i_1, \cdots, i_k\}$. The stablizer at this point is some copies of $\mathbb{G}_m$ with some copies of $\mathbb{G}_a$, in particular is connected. So there are no local system issues.

## 11.2 Multiplication Table for the Module

From general theory, we know $K(B \setminus X)$ is a $K(B \setminus G/B) \simeq \mathcal{H}_q(W)$-module. We wish to explicitly describe the module structure. Note it has a basis:

$$M_{w,S} = [i_{\mathcal{O}_{w,S}!}\overline{\mathbb{Q}_\ell}]$$

The key identity to use:

$$\mathrm{IC}_s * \mathcal{F} = \pi_s^* \pi_{s,*} \mathcal{F}[1](\frac{1}{2})$$

for each $\mathcal{F} \in D(B \setminus X)$, where $\pi_s = B \setminus X \to P_s \setminus X$. We need to compute $T_s \cdot M_{w,S}$, this requires us to understand $\pi_{s,*} i_{\mathcal{O}_{w,S}!}\overline{\mathbb{Q}_\ell}$, which is equivalent to understand the geometry of $P_s \mathcal{O}_{w,S}$. We know these have classification $U, G, T, N$.

1. Suppose $i, i+1 \notin S$, $\ell(w) < \ell(sw)$. Then:

$$P_s\mathcal{O}_{w,S} = \mathcal{O}(w,S) \cup \mathcal{O}(sw,S)$$

This has type $U$. It follows that $\pi_s|_{\mathcal{O}_{(w,S)}}$ is isomorphism to the target, hence:

$$T_s \cdot M_{w,S} = M_{sw,S}$$

2. Suppose $i \in S, i+1 \notin S$, $\ell(w) < \ell(sw)$. We compute:

$$P_s\mathcal{O}_{w,S} = \mathcal{O}_{w,S} \cup BsB\mathcal{O}_{w,S} = \mathcal{O}_{w,S} \cup \mathcal{O}_{sw,s(S)}$$

This has type $U$. It follows that $\pi_s|_{\mathcal{O}(w,S)}$ is isomorphism to the target, hence:

$$T_s \cdot M_{w,S} = M_{sw,s(S)}$$

3. Suppose $i+1 \in S, i \notin S, \ell(w) < \ell(sw)$. We compute:

$$P_s\mathcal{O}_{w,S} = \mathcal{O}_{w,S} \cup BsB \cdot (BwB, v)$$

We need to classify the $B_{sw}$-orbits of $BsBS$. But this is a local computation. We may assume $w = 1$, $S = \{i+1\}$:

$$BsB\begin{pmatrix}0\\1\end{pmatrix} = Bs\begin{pmatrix}*\\ \neq 0\end{pmatrix} = B\begin{pmatrix}\neq 0\\ *\end{pmatrix}$$

This contains both the orbit $\{e_i\}$ and $\{e_i + e_{i+1}\}$. Therefore:

$$P_s\mathcal{O}_{w,S} = \mathcal{O}_{w,S} \cup \mathcal{O}_{sw,s(S)} \cup \mathcal{O}_{sw,S\cup\{i\}}$$

This has type $T$ (the dense part is $\mathcal{O}_{sw,S\cup\{i\}}$). Hence:

$$T_s \cdot M_{w,S} = M_{sw,s(S)} + M_{sw,S\cup\{i\}}$$

4. Suppose $i, i+1 \in S$. Note in this case necessarily $\ell(w) > \ell(sw)$. This is the dense orbit in the type $T$ scenario as we seen in the previous part. Therefore:

$$\pi_{s,*}i_{\mathcal{O}_{w,S}!}\overline{\mathbb{Q}_\ell}[1](\frac{1}{2}) = H_c^*(\mathbb{G}_m[1](\frac{1}{2})) = (\overline{\mathbb{Q}_\ell}[-1]\oplus\overline{\mathbb{Q}_\ell}[-2](-1))[1](\frac{1}{2}) = \overline{\mathbb{Q}_\ell}(\frac{1}{2})\oplus\overline{\mathbb{Q}_\ell}[-1](-\frac{1}{2})$$

Hence:

$$\mathrm{IC}_s * i_{\mathcal{O}_{w,S}!}\overline{\mathbb{Q}_\ell} = \overline{\mathbb{Q}_\ell}(\frac{1}{2}) \oplus \overline{\mathbb{Q}_\ell}[-1](-\frac{1}{2})$$

Note $\mathrm{IC}_s$ is identified with $-C'_s = -q^{\frac{1}{2}}(T_s + 1)$. Hence:

$$-q^{\frac{1}{2}}(T_s+1)M_{w,S} = (q^{-\frac{1}{2}} - q^{\frac{1}{2}})(M_{w,S} + M_{w,S\setminus\{i\}} + M_{w,S\setminus\{i+1\}})$$

This rearranges to:

$$T_s \cdot M_{w,S} = (q-1)M_{w,S\setminus\{i\}} + (q-1)M_{w,S\setminus\{i+1\}} + (q-2)M_{w,S}$$

In summary, if $s=(i,i+1)$, the multiplication table is given by:

$$T_s\cdot M_{w,S}=\begin{cases} M_{sw,S}, & \text{if } i,i+1\notin S, \ell(w)<\ell(sw);\\ M_{sw,s(S)}, & \text{if } i\in S, i+1\notin S, \ell(w)<\ell(sw);\\ M_{sw,s(S)}+M_{sw,S\cup\{i\}}, & \text{if } i\notin S, i+1\in S, \ell(w)<\ell(sw);\\ (q-1)M_{w,S}+qM_{sw,S}, & \text{if } i,i+1\notin S, \ell(w)>\ell(sw);\\ (q-1)M_{w,S}+qM_{sw,s(S)}, & \text{if } i\notin S, i+1\in S, \ell(w)>\ell(sw);\\ M_{sw,s(S)}+M_{w,S\cup\{i+1\}}, & \text{if },i\in S, i+1\notin S, \ell(w)>\ell(sw);\\ (q-1)M_{w,S\setminus\{i\}}+(q-1)M_{w,S\setminus\{i+1\}} & \\ +(q-2)M_{w,S}, & \text{if } i,i+1\in S, \ell(w)>\ell(sw). \end{cases}$$

## 11.3 Decomposition when $q=1$

In this subsection, we decompose the above module $M_X$ into representations of $S_n$ for considering degeneration at $q=1$. Let's write down the multiplication table in this case first. For $s=(i,i+1)$

$$s\cdot M_{w,S}=\begin{cases} M_{sw,S}, & \text{if } i,i+1\notin S;\\ M_{sw,s(S)}, & \text{if } i\in S, i+1\notin S, \ell(w)<\ell(sw);\\ M_{sw,s(S)}, & \text{if } i\notin S, i+1\in S, \ell(w)>\ell(sw);\\ M_{sw,s(S)}+M_{sw,S\cup\{i\}}, & \text{if } i\notin S, i+1\in S, \ell(w)<\ell(sw);\\ M_{sw,s(S)}+M_{w,S\cup\{i+1\}}, & \text{if } i\in S, i+1\notin S, \ell(w)>\ell(sw);\\ -M_{w,S}, & \text{if } i,i+1\in S, \ell(w)>\ell(sw). \end{cases}$$

Note that $M_X$ has a filtration:

$$M_X^0\supset M_X^1\supset\cdots$$

where $M_X^i=\mathrm{Span}(M_{w,S}\mid |S|\geq i)$. Note this is a filtration of $S_n$-module from the above multiplication table. But $S_n$-modules are semisimple, hence:

$$M_X\simeq \mathrm{gr}M_X$$

Now $\mathrm{gr}M_X$ has multiplication table $s=(i,i+1)$ :

$$s\cdot M_{w,S}=\begin{cases} -M_{w,S}, & \text{if } i,i+1\in S, \ell(w)>\ell(sw);\\ M_{sw,s(S)}, & \text{otherwise.} \end{cases}$$

Therefore $S_n$ acts on $\mathrm{gr}M_X$ by permutation (up to some sign twists). The key observation is the that $w^{-1}(S)$ is an invariant, namely:

$$\mathrm{gr}M_X\simeq\bigoplus_{K\subset[n]}M_K,\quad M_K=\bigoplus_{(w,S):w^{-1}(S)=K}M_{w,S}$$

Let's now try to understand $M_K$. Start from any $M_{w,S}$, we choose $w'$ such that $w'(S)=\{1,2,\cdots,k\}$ for $k=|K|$. Then multiply $M_{w,K}$ by $w'$ we get:

$$M_{u,[k]}\in M_K,\quad [k]=\{1,2,\cdots,k\}\subset[n]$$

We may also assume $\ell(u)$ is minimal. In that case we must have $\ell(s_i u) > \ell(u)$ for $s_i = (i, i+1), i \geq k$. Note that $S_k = \mathrm{Sym}\{1, 2, \cdots, k\} \subset S_n$ acts on $M_{u,[k]}$ by sign, hence we have a well-define map:

$$\mathrm{Ind}_{S_k}^{S_n}(\mathsf{sgn}) \to M_K$$
$$w \otimes 1 \mapsto w \cdot M_{u,[k]}$$

We claim this map is surjective. Indeed, any $M_{w,S}$ can be obtained by $M_{u',[k]}$ for some $u'$, so it suffices to show we can get $u'$. But now:

$$u^{-1}([k]) = u'^{-1}([k]) = K \implies u^{-1}(i) = u'^{-1}(i), \forall i \leq k \implies u'u^{-1} \in \mathrm{Sym}(k+1, \cdots, k+n)$$

this is because $u^{-1}, u'^{-1}$ both descend on $[k]$. It follows that:

$$u'u^{-1} M_{u,[k]} = M_{u',[k]}$$

Hence the map is surjectivity. But now LHS has dimension $\frac{n!}{k!}$; for the RHS, we seek for all

$$(w, S) : w^{-1}(S) = K$$

Fix $S$, $w^{-1}$ must send $S$ to $K$, and $[n] \setminus S$ to $[n] \setminus K$ with prescribed ordering on $S \to K$. So there are $(n-k)!$ possibilities. Now we can vary $S$, and therefore dimension of RHS is:

$$\binom{n}{k} \times (n-k)! = \frac{n!}{k!}$$

This checks:

$$\mathrm{Ind}_{S_k}^{S_n}(\mathsf{sgn}) \to M_K$$

is an isomorphism. And:

$$M_X \simeq \bigoplus_{K \subset [n]} \mathrm{Ind}_{S_k}^{S_n}(\mathsf{sgn})$$

## 11.4 Some Classical Springer Theory

Let's work over $\mathbb{C}$. Recall in classical Springer theory, we have canonical isomorphisms:

$$\mathrm{End}(\mathrm{Spr}) \simeq H_{\mathrm{top}}(\mathrm{St}, \mathbb{C}) \simeq \mathbb{C}[W]$$

From decomposition theorem we know:

$$\mathrm{Spr} \simeq \bigoplus_{(\mathcal{O}, \mathcal{L})} \mathrm{IC}(\mathcal{O}, \mathcal{L}) \boxtimes V_{\mathcal{O},\mathcal{L}}$$

From this we obtain the irreducible $W$-modules:

$$E_{\mathcal{O},\mathcal{L}} = V_{\mathcal{O},\mathcal{L}}^* = \mathrm{Hom}(\mathrm{Spr}, \mathrm{IC}(\mathcal{O}, \mathcal{L}))$$

which is naturally a $\mathbb{C}[W] = \mathrm{End}(\mathrm{Spr})$-module. This relates to the Borel-Moore homology picture in the following way. Choose $e \in \mathcal{O}$ and consider Springer fibre $\mathcal{B}_e$. The local system $\mathcal{L}$ corresponds to $\chi \in \mathrm{Irr}_{\mathbb{C}}(A(e))$ for $A(e) = \pi_0(C_G(e))$. Then taking costalk of both sides of

$$\mathrm{Spr} \simeq \bigoplus_{(\mathcal{O},\mathcal{L})} \mathrm{IC}(\mathcal{O},\mathcal{L}) \boxtimes V_{\mathcal{O},\mathcal{L}}$$

at $e \in \mathcal{O}$ and then take cohomology in appropriate degree, yield an isomorphism of $A_e \times W$-modules:

$$H_{\mathrm{top}}(\mathcal{B}_e, \mathbb{C}) \simeq \bigoplus_{\chi \in A_e} \chi \boxtimes V_{\mathcal{O},\mathcal{L}}$$

Let's fix a choice of $T \subset B \subset G$. To relate the above discussion $K_0(D_B(\mathcal{B}))$, we need to use the **characteristic cycle** map. Let $\mu : \widetilde{\mathcal{N}} \to \mathcal{N}$ be the Springer resolution, and consider:

$$\Lambda = \mu^{-1}(\mathfrak{n}) = \bigcup_{w \in W} \overline{T^*_{\mathcal{O}(w)}\mathcal{B}}$$

is a Lagrangian in $\widetilde{\mathcal{N}}$. We have an identification:

$$G \times^B \Lambda \to \mathrm{St}$$

$$(g, x, \mathfrak{b}') \mapsto (gxg^{-1}, g\mathfrak{b}'g^{-1}, g\mathfrak{b}g^{-1})$$

where $\mathfrak{b}$ is the fixed Borel. In particular:

$$\mathrm{St} \to G/B$$

is a fibre bundle with fibre $\Lambda$. Now $G/B$ is proper smooth, it follows that:

$$H_{\mathrm{top}}(\Lambda, \mathbb{Z}) \simeq H_{\mathrm{top}}(\mathrm{St}, \mathbb{Z})$$

$$[C] \mapsto [G \times^B C]$$

Now we wish to define the characteristic cycle map as:

$$\mathrm{CC} : K_0(D_B(\mathcal{B}, \mathbb{Q})) \to H_{\mathrm{top}}(\Lambda, \mathbb{Z})$$

Implicitly, given an element in the LHS, we can use Riemann-Hilbert correspondence to turn it into a complex of holonomic $\mathcal{D}$-module, and we can take singular support there (which necessarily need to be a conormal bundle). One can show that:

$$\mathrm{CC}(\Delta_w) = [\Lambda_w] + (\text{ lower terms})$$

where $\Delta_w = j_{w!}\mathbb{C}$ is the standard object, and $\Lambda_w = \overline{T^*_{\mathcal{O}(w)}\mathcal{B}}$. From this formula one already see that CC is an isomorphism. In fact one can also show this is a ring homomorphism. Tensoring with $\mathbb{C}$ we get:

$$\mathrm{CC}_{\mathbb{C}} : K_0(D^b_B(\mathcal{B})) \otimes \mathbb{C} \to H_{\mathrm{top}}(\Lambda, \mathbb{C}) \simeq H_{\mathrm{top}}(\mathrm{St}, \mathbb{C})$$

## 11.5 Springer Theory for Spherical Varieties

We can extend the discussion by replace the flag variety $G/B$ by any spherical variety $X$ as follows. The $G$-action on $X$ induces a moment map:

$$\mu_{T^*X} : T^*X \to \mathfrak{g}^*$$

We will denote $\mathfrak{X} = T^*X$ to ease the notation. Now let:

$$\Lambda_{\mathfrak{X}} = \mu_{\mathfrak{X}}^{-1}(\mathfrak{b}^{\perp}) \subset \mathfrak{X}$$

be a Lagrangian.

**Definition 11.1.** The **$\mathfrak{X}$-Steinberg scheme** is defined to be:

$$\mathrm{St}_{\mathfrak{X}} = \mathfrak{X} \times_{\mathfrak{g}^*} \widetilde{\mathcal{N}}$$

So $\mathrm{St}_{\mathfrak{X}}$ fits into the Cartesian diagarm:

$$\begin{array}{ccccc} \mathrm{St}_{\mathfrak{X}} & \xrightarrow{\nu_{\mathfrak{X}}} & \widetilde{\mathcal{N}} & & \\ {\scriptstyle \Pi}\downarrow & & \downarrow{\scriptstyle \pi} & \searrow^{\pi_{\mathcal{N}}} & \\ \mathfrak{X} & \xrightarrow[\mu_{\mathfrak{X}}]{} & \mathfrak{g}^* & \xleftarrow{i_{\mathcal{N}}} & \mathcal{N} \end{array}$$

As before we have:

$$G \times^B \Lambda_{\mathfrak{X}} = \mathrm{St}_{\mathfrak{X}}$$

and thus:

$$H_{\mathrm{top}}(\Lambda_{\mathfrak{X}}, \mathbb{C}) \simeq H_{\mathrm{top}}(\mathrm{St}_{\mathfrak{X}}, \mathbb{C})$$

We also need a notion of **$\mathfrak{X}$-Springer sheaf**:

$$\mathrm{Spr}_{\mathfrak{X}} = i_{\mathcal{N}}^! \mu_{\mathfrak{X},*} \omega_{\mathfrak{X}}[-2\dim \mathrm{St}_{\mathfrak{X}} + \dim \mathcal{N}] \in D_G(\mathcal{N})$$

**Lemma 11.2.** *There is a natural isomorphism:*

$$\mathrm{Hom}_{D_G(\mathcal{N})}(\mathrm{Spr}, \mathrm{Spr}_{\mathfrak{X}}) \simeq H_{\mathrm{top}}(\mathrm{St}_{\mathfrak{X}}, \mathbb{C})$$

*endowing $H_{\mathrm{top}}(\mathrm{St}_{\mathfrak{X}}, \mathbb{C})$ the structure of a $W$-module.*

**Remark.** *An equivalent way to describe the $W$-action is the following. We have $W$ acting on the classical Springer sheaf $\mathrm{Spr} \in D_G(\mathcal{N})$. Now we have:*

$$\Pi_* \mathbb{C}_{\mathrm{St}_{\mathfrak{X}}}[\dim \widetilde{\mathcal{N}}] \simeq \mu_{\mathfrak{X}}^* i_{\mathcal{N}!} \mathrm{Spr} \in D_G(\mathfrak{X})$$

*with a $W$-action. Since $\Pi$ is proper, we thus have a natural action on:*

$$H_c^*(\mathrm{St}_{\mathfrak{X}}, \mathbb{C}) \simeq H_*(\mathrm{St}_{\mathfrak{X}}, \mathbb{C})$$

As before, we can define a cycle class map:

$$\mathrm{CC}_{\mathbb{C}} : K_0(D_B^b(X)) \otimes \mathbb{C} \to H_{\mathrm{top}}(\mathrm{St}_{\mathfrak{x}}, \mathbb{C})$$

This will be an $W$-equivariant isomorphism after tensoring with the sign representation of $W$, namely:

$$\mathrm{CC}_{\mathbb{C}}(w \cdot [\mathcal{F}]) = (-1)^{\ell(w)} w \cdot \mathrm{CC}_{\mathbb{C}}(\mathcal{F})$$

Finally, let's understand how to decompose $H_{\mathrm{top}}(\mathrm{St}_{\mathfrak{x}}, \mathbb{C})$ into irreducibles. Let $e$ be a nilpotent element, we define:

$$\mathfrak{X}_e = \mu_{\mathfrak{x}}^{-1}(e)$$

Then $H_{\mathrm{top}}(\mathfrak{X}_e, \mathbb{C})$ carries a natural $A_e = \pi_0(C_G(e))$-action. For each $\chi \in \mathrm{Irr}(A_e)$, we define:

$$V_{\mathcal{O},\mathcal{L}}(\mathfrak{X}) = V_{e,\chi}(\mathfrak{X}) = \mathrm{Hom}_{A_e}(\chi, H_{\mathrm{top}}(\mathfrak{X}_e, \mathbb{C}))$$

**Definition 11.3.** 1. A nilpotent orbit $\mathcal{O} \subset \mathcal{N}$ is called $\mathfrak{X}$-relevant if:

$$\dim \mu_{\mathfrak{x}}^{-1}(\mathcal{O}) = \dim \mathrm{St}_{\mathfrak{x}} + \frac{1}{2}\dim \mathcal{O} - \frac{1}{2}\dim \widetilde{\mathcal{N}}$$

Let $\mathcal{R}_G(\mathfrak{X})$ be the set of relevant nilpotent $G$-orbits. This is saying the dimension of $\mathfrak{X}_e$ is "as expected".

2. A pair $(\mathcal{O}, \mathcal{L})$ is called $\mathfrak{X}$-relevant pair if $\mathcal{O} \in \mathcal{R}_G(\mathfrak{X})$ and $V_{\mathcal{O},\mathcal{L}} \neq 0$. We denote the set of $\mathfrak{X}$-relevant pairs to be $\widehat{\mathcal{R}}_G(\mathfrak{X})$

**Remark.** *We always have:*

$$\dim \mu_{\mathfrak{x}}^{-1}(\mathcal{O}) \leq \dim \mathrm{St}_{\mathfrak{x}} + \frac{1}{2}\dim \mathcal{O} - \frac{1}{2}\dim \widetilde{\mathcal{N}}$$

**Theorem 11.4.** *We have a decomposition of $H_{\mathrm{top}}(\mathrm{St}_{\mathfrak{x}}, \mathbb{C})$ as a $W$-module:*

$$H_{\mathrm{top}}(\mathrm{St}_{\mathfrak{x}}, \mathbb{C}) \simeq \bigoplus_{(\mathcal{O},\mathcal{L}) \in \widehat{\mathcal{R}}_G(\mathfrak{x})} E_{\mathcal{O},\mathcal{L}} \otimes V_{\mathcal{O},\mathcal{L}}(\mathfrak{X})$$

*where $E_{\mathcal{O},\mathcal{L}}$ are the Springer representations.*

Let's give a sketch for this theorem. Some computation involving six functors give:

**Lemma 11.5.** $\mathrm{Spr}_{\mathfrak{x}} \in {}^p D_G^b(\mathcal{N})^{\geq 0}$

As a consequence, we can define:

$$\mathrm{Spr}_{\mathfrak{x}}^{\diamond} \coloneqq {}^p\tau_{\leq 0}\mathrm{Spr}_{\mathfrak{x}} \in \mathrm{Perv}_G(\mathcal{N})$$

be the truncation with respect to perverse cohomology. Now we can decompose:

$$\mathrm{Spr}_{\mathfrak{x}}^{\diamond} = \bigoplus \mathrm{IC}(\mathcal{O}, \mathcal{L}) \otimes M_{\mathcal{O},\mathcal{L}}$$

for $M_{\mathcal{O},\mathcal{L}}$ the multiplicity spaces. So that:

$$H_{\mathrm{top}}(\mathrm{St}_{\mathfrak{x}}, \mathbb{C}) \simeq \mathrm{Hom}_{D_G^b(\mathcal{N})}(\mathrm{Spr}, \mathrm{Spr}_{\mathfrak{x}}) \simeq \mathrm{Hom}_{\mathrm{Perv}_G(\mathcal{N})}(\mathrm{Spr}, \mathrm{Spr}_{\mathfrak{x}}^{\diamond}) = \bigoplus E_{\mathcal{O},\mathcal{L}} \otimes M_{\mathcal{O},\mathcal{L}}$$

Therefore it suffices to show:

$$M_{\mathcal{O},\mathcal{L}} = \begin{cases} V_{\mathcal{O},\mathcal{L}}(\mathfrak{X}), & \text{if } (\mathcal{O},\mathcal{L}) \in \widehat{\mathcal{R}}_G(\mathfrak{X}); \\ 0, & \text{otherwise.} \end{cases}$$

## 11.6 Compute Springer Fibres

Let's now conitnue with our previous example $X = \mathcal{B} \times \mathbb{C}^n$, and $\mathfrak{X} = \widetilde{\mathcal{N}} \times T^*\mathbb{C}^n$ . By the generality from last section, we know:

$$H_{\text{top}}(\text{St}_{\mathfrak{X}}, \mathbb{C}) = \bigoplus_{e \in \mathcal{R}_G(X)} E_e \otimes H_{\text{top}}(\mathfrak{X}_e, \mathbb{C})$$

So we need to understand the set $\mathcal{R}_G(X)$ and the Springer fibres $\mathfrak{X}_e$. The moment map reads:

$$\mu : \widetilde{\mathcal{N}} \times T^*\mathbb{C}^n \to \mathfrak{g}$$
$$(x, \mathfrak{b}, v, p) \mapsto x + vp^t$$

where we identify $\mathfrak{g}^* \simeq \mathfrak{g}$ via the trace form, identify $T^*(\mathcal{B} \times \mathbb{C}^n) \simeq T^*\mathcal{B} \times T^*\mathbb{C}^n$ as symplectic manifolds, and identify $\mathbb{C}^n$ with $(\mathbb{C}^n)^*$ using the standard Euclidean inner product on $\mathbb{C}^n$. Therefore, the Springer fibre:

$$\mathfrak{X}_e = \{(\mathfrak{b}, v, p) \mid e - vp^t \in \mathfrak{n}\}$$

Consider the map:

$$\mathfrak{X}_e \to \mathbb{C}^n \times \mathbb{C}^n$$

Then if $(v, p)$ lies in the target, the fibre will be $\mathcal{B}_{e-vp^t}$. Let's call the image $S_e \subset \mathbb{C}^n \times \mathbb{C}^n$. We require:

$$x^n = \det(xI - e + vp^t) = \det(xI - e)(p^t(xI - e)^{-1}v) = x^n \cdot p^t(\frac{1}{x} + \frac{e}{x^2} + \cdots)v$$

This forces: $p^t e^k v = 0$ for all $0 \leq k \leq n-1$. Hence:

$$S_e = \{(v, p) \mid p^t e^k v = 0, \forall k\}$$

**Hope**: Suppose nilpotent element $e$ corresponds to partition $\lambda$, then $\mathfrak{X}_e$ is equidimensional, with number of irreducible components equal:

$$\sum_{i=0}^{n} \binom{n}{i} [\text{Ind}_{S_i \times (S_1)^{n-i}}^{S_n} \mathsf{triv} : V_\lambda] = \sum_{i=0}^{n} \binom{n}{i} K_{\lambda,(i,1^{n-i})}$$

And the dimension of the $\mathfrak{X}_e$ equals:

$$n + \dim \mathcal{B}_e$$

**Example.** *If $e = 0$. Then:*

$$\mathfrak{X}_e = \mathbb{C}^n \times \{0\} \times \mathcal{B} \cup \{0\} \times \mathbb{C}^n \times \mathcal{B} \cup \{(v, p) \mid p^t v = 0\} \times \mathcal{B}_{(2,1^{n-2})}$$

*From hook lenght formula, we get $1 + 1 + n - 1 = n + 1$ irreducible components. Note:*

$$\dim \mathcal{B} + n = \frac{n(n+1)}{2}$$

*Recall that the dimension of Springer fibre is given by:*

$$\dim \mathcal{B}_\lambda = \sum (i-1)\lambda_i$$

*It follows that:*

$$\dim \mathcal{B}_{(2,1^{n-2})} = \frac{(n-1)(n-2)}{2} \implies \dim\{(v,p) \mid p^t v = 0\} \times \mathcal{B}_{(2,1^{n-2})} = \frac{n(n+1)}{2}$$

*So $\mathfrak{X}_e$ is equidimensional, with $n+1$ irreducible components. Note in this case, we have:*

$$n+1 = \binom{n}{0} K_{(1^n),(1^n)} + \binom{n}{1} K_{(1^n),(1^n)}$$

*Confirming our hope.*

**Example.** *Let $e$ be regular nilpotent. In this case $S_e$ is defined by $n$ equations, hence has dimension $n$. Now $\mathfrak{X}_e \to S_e$ has generic fibre a single point. Hence once we show $\mathfrak{X}_e$ is equidimensional, we will find:*

$$\dim \mathfrak{X}_e = \dim S_e = n$$

*It's a straightforward computation to see that:*

$$S_e = \bigcup_{i=0}^{n} \{(v_1, \cdots, v_i, 0, \cdots) \times (0, 0, \cdots, p_{i+1}, p_{i+2}, \cdots)\}$$

*Call the $i$-th piece $S_e^i$. For $(v,p) \in S_e^i$, we can write:*

$$e - vp^t = \begin{pmatrix} J_i & X \\ 0 & J_{n-i} \end{pmatrix}$$

*where* $X = \begin{pmatrix} -v_1p_{i+1} & -v_1p_{i+2} & \cdots & -v_1p_n \\ -v_2p_{i+1} & -v_2p_{i+2} & \cdots & -v_2p_n \\ \cdots & \cdots & \cdots & \cdots \\ 1 - v_ip_{i+1} & -v_ip_{i+2} & \cdots & -v_ip_n \end{pmatrix}$ *If $v_ip_{i+1} \neq 0$, then $e - vp^t$ is regular nilpotent (the generic case). Otherwise, $e - vp^t$ must has Jordan type $(n-k,k)$ for some $k \geq \frac{n}{2}$. We can explicitly compute $\sum_{a+b=N} J_i^a X J_{n-i}^b$ and find the follwing. Let*

$$f_1 = v_ip_{i+1} - 1, f_2 = v_ip_{i+2} + v_{i-1}p_{i+1}, f_3 = v_ip_{i+3} + 2v_{i-1}p_{i+2} + v_{i-2}p_{i+1}, \cdots$$

*with general term:*

$$f_j = v_ip_{i+j} + \binom{j-1}{1} v_{i-1}p_{i+j-1} + \binom{j-1}{2} v_{i-2}p_{i+j-2} + \cdots + v_{i-j+1}p_{i+1}$$

*And $e - vp^t$ has Jordan type $(n-k,k)$ is defined by:*

$$f_1 = 0, \cdots, f_k = 0, f_{k+1} \neq 0$$

*Therefore, the preimage of $S_e^i$ (say $i \leq \frac{n}{2}$) equals:*

$$\bigcup_{k=0}^{i} S_e^{i,k}$$

*where $S_e^{i,k} \to \{f_1 = f_2 = \cdots = f_k = 0\}$ is a fibre bundle with fibre $\mathcal{B}_{(n-k,k)}$. Therefore, it has dimension $(n-k)+k = n$. This checks $\mathfrak{X}_e$ is equidimensional. Moreover, $S_e^{i,k}$ has $\binom{n+1}{k}(1-\frac{2k}{n+1})$ irreducible components. Therefore the number of irreducible components of $\mathfrak{X}_e$ equals:*

$$\sum_{i=0}^{n} \sum_{k=0}^{\min\{i,n-i\}} \binom{n+1}{k}(1-\frac{2k}{n+1}) = 2^n$$

*as desired.*

Let's now show that $\mathfrak{X}_e$ is indeed equidimensional using symplectic methods.

**Theorem 11.6.** *The mirabolic Springer fibre $\mathfrak{X}_e$ is equidimensional of dimension $n+\dim \mathcal{B}_e$.*

*Proof.* As in classical Springer theory, we have the diagram:

$$\begin{array}{ccccc} \mathfrak{X}_e & \hookrightarrow & \widetilde{\mathcal{O}} & \hookrightarrow & T^*X = \mathfrak{X} \\ \downarrow & & \downarrow & & \downarrow \mu \\ \{e\} & \hookrightarrow & \mathcal{O} & \hookrightarrow & \mathfrak{g} \end{array} \qquad \begin{array}{c} (x, \mathfrak{b}, v, p) \\ \downarrow \\ x + vp^t \end{array}$$

where $\mathcal{O}$ is a nilpotent orbit containing $e$. The goal is to show that $\widetilde{\mathcal{O}}$ is equidiemnsional with:

$$\dim \widetilde{\mathcal{O}} = \dim \mathfrak{X}_e + \dim \mathcal{O} = n + \dim \mathcal{B}_e + \dim \mathcal{O} = n + \dim G/B_0 + \frac{1}{2}\dim \mathcal{O}$$

Let's look at:

$$\begin{array}{ccc} F & & \mathfrak{n} \times T^*V \\ \downarrow & & \downarrow \\ \widetilde{\mathcal{O}} & \hookrightarrow & T^*X \\ \downarrow & & \downarrow \\ G/B_0 & \xrightarrow{=} & G/B_0 \end{array}$$

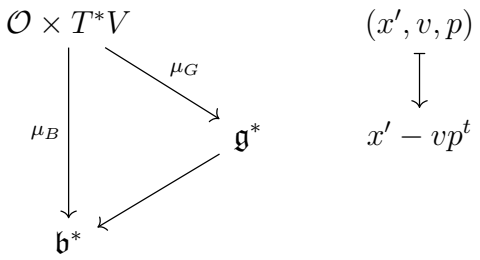
Since the $G$ acts transitively on the base, it follows that LHS is also a fibre bundle, with fibre at $\mathfrak{b}_0$ equals:

$$F = \{(x, v, p) \mid x \in \mathfrak{n}_0, x + vp^t \in \mathcal{O}\}$$

Note we can view $F$ as a subspace of $\mathcal{O} \times T^*V$ via:

$$(x, v, p) \mapsto (x + vp^t, v, p) = (x', v, p)$$

We claim that $F \subset \mathcal{O} \times T^*V$ is a Lagrangian, and theirfore the result follows. Indeed, let's look at the $B$-action on $\mathcal{O} \times T^*V$, from which we get the moment map:




It is clear that $\mu_B^{-1}(0) = F$, and therfore $F$ is coisotropic (it is Hamiltonian reduction of a solvable group action).

It remains to show that $\dim F \leq \frac{1}{2}\dim \mathcal{O} + n$. But note we have diagram:

$$\begin{array}{ccccc} & & F & & \\ & & \downarrow & & \\ \mathfrak{X}_e & \longrightarrow & \widetilde{\mathcal{O}} & \longrightarrow & \mathcal{O} \\ & & \downarrow & & \\ & & G/B_0 & & \end{array}$$

So that:

$$\dim F = \dim \mathfrak{X}_e + \dim \mathcal{O} - \dim G/B_0$$

So we only need to bound the dimension of $\mathfrak{X}_e$. But this is again acheived using Steinberg variety. Note that if:

$$\mu_X : \mathfrak{X} \to \mathfrak{g}, \quad \mu_{\mathcal{N}} : \widetilde{\mathcal{N}} \to \mathcal{N}$$

be the two moment maps, then:

$$\mathrm{St}_{\mathcal{O}} = \mu_X^{-1}(\mathcal{O}) \times_{\mathcal{O}} \mu_{\mathcal{N}}^{-1}(\mathcal{O})$$

which sits inside a fibre bundle:

$$\mathfrak{X}_e \times \mathcal{B}_e \to \mathrm{St}_{\mathcal{O}} \to \mathcal{O}$$

But $\mathrm{St}_{\mathcal{O}} \subset \mathrm{St}_{\mathfrak{X}}$ hence has dimension $\leq \dim \mathrm{St}_{\mathfrak{X}} = 2\dim G/B + n$. It follows that:

$$\dim \mathfrak{X}_e + \dim \mathcal{B}_e + \dim \mathcal{O} \leq 2\dim G/B + n$$

Rearranges we get our desired bound. □

**Remark.** *We used the fact that $X$ is a spherical variety, for using $\Lambda_{\mathfrak{X}}$ is a Lagrangian.*

## 11.7 Extra Symmetry: Two-sided Cells

Let $G = \mathrm{GL}_n = \mathrm{GL}(V)$. Let's consider $G \times G^{\mathrm{op}}$-variety:

$$Y = G \times V$$

For first component acts diagonally (on the left) and second component only acts on the first part (on the right). From this action we get a $K(B \setminus G/B) \times K(B \setminus G/B)$-bimodule:

$$K(B \setminus G \times V/B) \simeq K(B \setminus G/B \times V)$$

This is saying that our original left $\mathbb{C}[S_n]$ module $M_X$ in fact has a commuting right $\mathbb{C}[S_n]$-action. Again we have two equivalent description of this module: Kazhdan-Lusztig theory and Springer theory.

Let's start with the Kazhdan-Lusztig description. Let's start by discussing some general patterns we've seen in our previous example. Let $X$ be a right $G$-spherical variety, and let $\mathcal{O}$ be a $B$-orbit on $X$, so that $M_X$ has a basis given by $\{M_{\mathcal{O}} = [\mathcal{F}_{\mathcal{O}}] = [i_{\mathcal{O}!}\overline{\mathbb{Q}_\ell}]\}$. Now that the Hecke algebra is generated by:

$$\{\mathscr{C}_s : s \text{ simple reflection}\}$$

for $\mathscr{C}_s$ the un-normalized simple perverse sheaves (namely $[\mathscr{C}_s] \mapsto T_s + 1$ in the KL isomorphism). We know that:

$$\mathcal{F}_{\mathcal{O}} * \mathscr{C}_s = \pi_s^* \pi_{s,*} \mathcal{F}_{\mathcal{O}}$$

for $\pi_s : X/B \to X/P_s$. The computation from previous section tells us:

1. If $P_s \times^B \mathcal{O} \simeq P_s\mathcal{O}$, then if $P_s\mathcal{O} = \mathcal{O} \cup \mathcal{O}'$ (this happens in type U) or $P_s\mathcal{O} = \mathcal{O} \cup \mathcal{O}' \cup \mathcal{O}''$ (this happens in type T) then:

$$\mathcal{F}_{\mathcal{O}} * \mathscr{C}_s = \mathcal{F}_{\mathcal{O}} + \mathcal{F}_{\mathcal{O}'} \text{ or } \mathcal{F}_{\mathcal{O}} + \mathcal{F}_{\mathcal{O}'} + \mathcal{F}_{\mathcal{O}''}$$

$$\implies M_{\mathcal{O}} \cdot T_s = M_{\mathcal{O}'} \text{ or } M_{\mathcal{O}'} + M_{\mathcal{O}''}$$

2. If I'm in the setting of type $T$ and $\mathcal{O} \subset P_s\mathcal{O}$ is open dense, and assume $q = 1$, then:

$$\mathcal{F}_{\mathcal{O}} * \mathscr{C}_s = 0 \implies M_{\mathcal{O}} \cdot T_s = -M_{\mathcal{O}}$$

Let's specialize in our setting: $X = B \setminus G \times V$. The right $B$ orbits are classified by:

$$\{M_{w,S} \mid w \in W, S \subset [n], (i < j) \in S \implies w^{-1}(i) > w^{-1}(j)\}$$

where $(w, e_S) \in M_{w,S}$ for $e_S = \sum_{i \in S} e_i$. We need to describe:

$$(w, e_S) \cdot P_s = (w \cdot P_s, e_S)$$

It really reduces to a $\mathrm{SL}_2$-computation. Let's assume $P_s = \mathrm{GL}_2$, then $M_X$ has a basis:

$$M_{1,\emptyset} \quad M_{1,\{1\}} \quad M_{1,\{2\}} \quad M_{s,\emptyset} \quad M_{s,\{1\}} \quad M_{s,\{2\}} \quad M_{s,\{1,2\}}$$

We compute that:

$$M_{1,\emptyset} \xrightarrow{T_s} M_{s,\emptyset}$$

$$M_{1,\{1\}} \xrightarrow{T_s} M_{s,\{1\}}$$

And $M_{1,\{2\}}, M_{s,\{2\}}, M_{s,\{1,2\}}$ are glued together in the form of type $T$, with $M_{s,\{1,2\}}$ open dense. Therefore:

$$M_{1,\{2\}} \xrightarrow{T_s} M_{s,\{2\}} + M_{s,\{1,2\}}$$

$$M_{s,\{2\}} \xrightarrow{T_s} M_{1,\{2\}} + M_{s,\{1,2\}}$$

$$M_{s,\{1,2\}} \xrightarrow{T_s} M_{s,\{1,2\}}$$

This shows that:

$$M_{w,S} \cdot T_s = \begin{cases} M_{ws,S}, & \text{if } w(i), w(i+1) \notin S; \\ M_{ws,S}, & \text{if } \ell(w) < \ell(ws), w(i) \in S, w(i+1) \notin S; \\ M_{ws,S}, & \text{if } \ell(w) > \ell(ws), w(i) \notin S, w(i+1) \in S; \\ M_{ws,S} + M_{ws,S\cup\{w(i)\}}, & \text{if } \ell(w) < \ell(ws), w(i) \notin S, w(i+1) \in S; \\ M_{ws,S} + M_{ws,S\cup w(i+1)}, & \text{if } \ell(w) > \ell(ws), w(i) \in S, w(i+1) \notin S; \\ -M_{w,S}, & \text{if } w(i), w(i+1) \in S. \end{cases}$$

Now our previous filtration on cardinality of $S$ still preserves this right action. Therefore after taking associated graded we get the module:

$$s \cdot M_{w,S} = \begin{cases} -M_{w,S}, & \text{if } i, i+1 \in S; \\ M_{sw,s(S)}, & \text{otherwise.} \end{cases}$$

$$M_{w,S} \cdot s = \begin{cases} -M_{w,S}, & \text{if } w(i), w(i+1) \in S; \\ M_{ws,S}, & \text{otherwise.} \end{cases}$$

Let $M_k = \bigoplus_{|S|=k} \mathbb{C} \cdot M_{w,S}$. Then:

$$M_X = \bigoplus_k M_k$$

with $S_n \times S_n$ acts on each piece by permutation. Let's try to describe the bimodule structure, and for this purpose we need a reparametrization. Note $(w, S)$ is recorded in the following data:

$$(S, K, \sigma)$$

where $K = w^{-1}(S)$ and $\sigma : [n] \setminus K \to [n] \setminus S$ is the restriction of $w$. This is because the order of $w : K \to S$ is prescribed. Using this parametrization:

$$s \cdot M_{S,K,\sigma} = \begin{cases} -M_{S,K,\sigma}, & \text{if } i, i+1 \in S; \\ M_{s(S),K,s\sigma}, & \text{otherwise.} \end{cases}$$

$$M_{S,K,\sigma} \cdot s = \begin{cases} -M_{S,K,\sigma}, & \text{if } i, i+1 \in K; \\ M_{S,s(K),\sigma s}, & \text{otherwise.} \end{cases}$$

Let's view both actions as left actions. If start from the element:

$$M_{[k],[k],\mathrm{id}}$$

then $S_k \times 1, 1 \times S_k$ acts as $\mathsf{sgn}$ and $\Delta(1 \times S_{n-k})$ acts as $\mathsf{triv}$. So we have a well-defined map:

$$\mathrm{Ind}_{(S_k \times S_k) \times \Delta(1 \times S_{n-k})}^{S_n \times S_n}(\mathsf{sgn} \boxtimes \mathsf{triv}) \to M_k$$

Surjectivity is clear. Let's also compute dimension on both sides:

$$\dim(\mathrm{LHS}) = \frac{(n!)^2}{(k!)^2(n-k)!}$$

$$\dim(\mathrm{RHS}) = \binom{n}{k} \cdot \frac{n!}{k!} = \frac{(n!)^2}{(k!)^2(n-k)!}$$

Hence we obtain the isomorphism. Note if we want to transfer to the Springer setting we need to apply a $\mathsf{sgn}$ twist and we're getting precisely:

$$\bigoplus_k \mathrm{Ind}_{(S_k \times S_k) \times \Delta(1 \times S_{n-k})}^{S_n \times S_n}(\mathsf{triv} \boxtimes \mathsf{triv})$$

Let's try to decompose this into irreducibles. Let denote $\Gamma = S_n \times S_n, P = S_k \times S_k, Q = S_{n-k} \times S_{n-k}, R = \Delta(S_{n-k}) \subset Q$. The multiplicity $m_{\lambda,\mu}^k$ of $V_\lambda \boxtimes V_\mu$ appearaing in $M_k$ equals:

$$\dim \mathrm{Hom}_\Gamma(\mathrm{Ind}_{(S_k \times S_k) \times \Delta(1 \times S_{n-k})}^{S_n \times S_n}(\mathsf{triv} \boxtimes \mathsf{triv}), V_\lambda \boxtimes V_\mu) = \dim \mathrm{Hom}_{P \times R}(\mathsf{triv}, \mathrm{Res}_{P \times R}^\Gamma V_\lambda \boxtimes V_\mu)$$

First note:

$$\mathrm{Res}_{P \times Q}^\Gamma V_\lambda \boxtimes V_\mu = \bigoplus_{\alpha \vdash k, \beta \vdash n-k, \gamma \vdash k, \theta \vdash n-k} (V_\alpha \boxtimes V_\beta)^{\oplus c_{\alpha,\beta}^\lambda} \boxtimes (V_\gamma \boxtimes V_\theta)^{\oplus c_{\gamma,\theta}^\mu}$$

Therefore:

$$m_{\lambda,\mu}^k = \sum_{\alpha,\beta,\gamma,\theta} c_{\alpha,\beta}^\lambda \cdot c_{\gamma,\theta}^\mu \dim \mathrm{Hom}_{P \times R}(\mathrm{Res}_{P \times R}^{P \times Q}(V_\alpha \boxtimes V_\gamma \boxtimes V_\beta \boxtimes V_\theta), \mathsf{triv})$$

Note the inner space equals:

$$\mathrm{Hom}_P(V_\alpha \boxtimes V_\gamma, \mathsf{triv}) \times \mathrm{Hom}_R(\mathrm{Res}_R^Q(V_\beta \boxtimes V_\theta), \mathsf{triv})$$

The first space is nonzer iff $\alpha = \gamma = (k)$ (and in that case it is one dimensional); the second space equals:

$$\mathrm{Hom}_Q(V_\beta \boxtimes V_\theta, \mathbb{C}[Q])$$

which is one dimensional iff $\beta = \theta$. Therefore, we have:

$$m_{\lambda,\mu}^k = \sum_{\beta \vdash n-k} c_{(k),\beta}^\lambda \cdot c_{(k),\beta}^\mu$$

Note that:

$$c^{\lambda}_{(k),\beta} \cdot c^{\mu}_{(k),\beta} = \begin{cases} 1, & \text{if } \lambda/\beta = \mu/\beta \text{ are } k \text{ horizontal strips;} \\ 0, & \text{otherwise.} \end{cases}$$

Therefore $m^k_{\lambda,\mu}$ counts $\beta \vdash n-k$ that is obtained by removing a $k$-horizontal strip from both $\lambda$ and $\mu$. Therefore:

$$M_X \simeq \bigoplus_{\lambda,\mu} (V_\lambda \boxtimes V_\mu)^{\oplus m_{\lambda,\mu}}$$

where $m_{\lambda,\mu}$ counts all partitions $\beta$ which can be obtained by removing a horizontal strip from both $\lambda$ and $\mu$.

Let's now try the Springer theory approach. We need to be careful of left and right. Indeed we have identification:

$$G \times \mathfrak{g}^* \simeq T^*G$$
$$(g, \alpha) \mapsto (g, (dL_{g^{-1}})^*(\alpha))$$

The left $G$-action on $T^*G$ is given by:

$$g \cdot (h, \lambda) = (gh, (dL_{g^{-1}})^*(\lambda))$$

which translates to

$$g \cdot (h, \alpha) = (gh, \alpha)$$

using the equivariant identification. Therefore the moment map is given by:

$$\mu_1 : G \times \mathfrak{g}^* \to \mathfrak{g}^*$$

and for $x \in \mathfrak{g}$

$$\mu_1(g, \alpha)(x) = \langle dL_g(x), (dL_{g^{-1}})^*(\alpha) \rangle = \langle x, \alpha \rangle \implies \mu_1(g, \alpha) = \alpha$$

On ther other hand, the right $G$-action on $T^*G$ is given by:

$$(h, \lambda) \cdot g = (hg, (dR_{g^{-1}})^* \lambda)$$

which translates to:

$$(h, \alpha) \cdot g = (hg, (dL_g \circ dR_{g^{-1}})^* \alpha) = (hg, \mathrm{Ad}^*_g(\alpha))$$

Therefore the moment is given by:

$$\mu_2 : G \times \mathfrak{g}^* \to \mathfrak{g}^*$$

and for $x \in \mathfrak{g}$

$$\mu_2(g, \alpha)(x) = \langle dR_g(x), (dL_{g^{-1}})^*(\alpha) \rangle = \langle dL_{g^{-1}} \circ dR_g(x), \alpha \rangle = \langle \mathrm{Ad}_{g^{-1}}(x), \alpha \rangle$$

Hence: $\mu_2(g, \alpha) = \mathrm{Ad}^*_{g^{-1}}(\alpha)$. Now if $Y = G \times V$, and use Killing form to identify $\mathfrak{g} \simeq \mathfrak{g}^*$, the moment map:

$$\mu_Y : T^*Y \simeq G \times \mathfrak{g} \times V \times V^* \to \mathfrak{g} \times \mathfrak{g}$$

is given by:

$$(g, x, v, p) \mapsto (x + vp^t, \mathrm{Ad}_g(x))$$

In this case the Springer fibre is:

$$\mathfrak{Y}_{e,e'} = \{(g, x, v, p) \mid x + vp^t = e, gxg^{-1} = e'\}$$

Which sits inside the fibre bundle:

$$\mathfrak{Y}_{e,e'} \to \{(g, x, v, p) \mid x + vp^t \in \mathcal{O}_e, x \in \mathcal{O}_{e'}\} \to \mathcal{O}_e \times \mathcal{O}_{e'}$$

We can also rewrite the middle space as:

$$G \times Z_{e,e'}, \quad Z_{e,e'} = \{(x, v, p) \mid x + vp^t \in \mathcal{O}_e, x \in \mathcal{O}_{e'}\}$$

Some linear algebra enables us to show that $Z_{e,e'} \neq \emptyset$ precisely when $e, e'$ corresponds to **interlacing partitions**. Note $\mathfrak{Y}_{e,e'}$ will be a relevant orbit iff:

$$\dim Z_{e,e'} = n + \frac{1}{2}\dim \mathcal{O}_e + \frac{1}{2}\dim \mathcal{O}_{e'}$$

But consider the diagonal $G$-action on $\mathcal{O}_e \times \mathcal{O}_{e'} \times T^*V$, the moment map is given by:

$$\mu' : \mathcal{O}_e \times \mathcal{O}_{e'} \times T^*V \to \mathfrak{g}$$

$$\mu'(x, y, v, p) = x - y + vp^t$$

and therefore $\mu'^{-1}(0) = Z_{e,e'}$. But $\mu'$ is **equivariant**, so that $Z_{e,e'} \subset \mathcal{O}_e \times \mathcal{O}_{e'} \times T^*V$ is coisotropic. To see it's actually a Lagrangian, recall that we defined:

$$F = \{(x, v, p) \mid x \in \mathfrak{n}_0, x + vp^t \in \mathcal{O}_e\}$$

It has a subspace:

$$F_{\mathcal{O}_{e'} \cap \mathfrak{n}_0} = \{(x, v, p) \mid x \in \mathfrak{n}_0 \cap \mathcal{O}_{e'}, x + vp^t \in \mathcal{O}_e\}$$

It sits inside the fibre bundle:

$$F_{e'} = \{(v, p) \mid e' + vp^t \in \mathcal{O}_e\} \to F_{\mathcal{O}_{e'} \cap \mathfrak{n}_0} \to \mathfrak{n}_\mathfrak{o} \cap \mathcal{O}_{e'}$$

where the second map is $(x, v, p) \mapsto x$. Moreover, $Z_{e,e'}$ sits inside fibre bundle:

$$F_{e'} \to Z_{e,e'} \to \mathcal{O}_{e'}$$

where the second map is again $(x, v, p) \mapsto x$. It follows that:

$$\dim(Z_{e,e'}) = \dim \mathcal{O}_{e'} - \dim(\mathfrak{n}_0 \cap \mathcal{O}_{e'}) + \dim F_{\mathcal{O}_{e'} \cap \mathfrak{n}_0}$$

But $\mathfrak{n}_0 \cap \mathcal{O}_{e'} \subset \mathcal{O}_{e'}$ is a Lagrangian, and

$$\dim F_{\mathcal{O}_{e'} \cap \mathfrak{n}_0} \leq \dim F = \frac{1}{2}\dim \mathcal{O}_e + n$$

We deduce that:

$$\dim(Z_{e.e'}) \leq n + \frac{1}{2}\dim \mathcal{O}_e + \frac{1}{2}\dim \mathcal{O}_{e'}$$

as desired.

**Remark.** *As a byproduct, we actually proved that:*

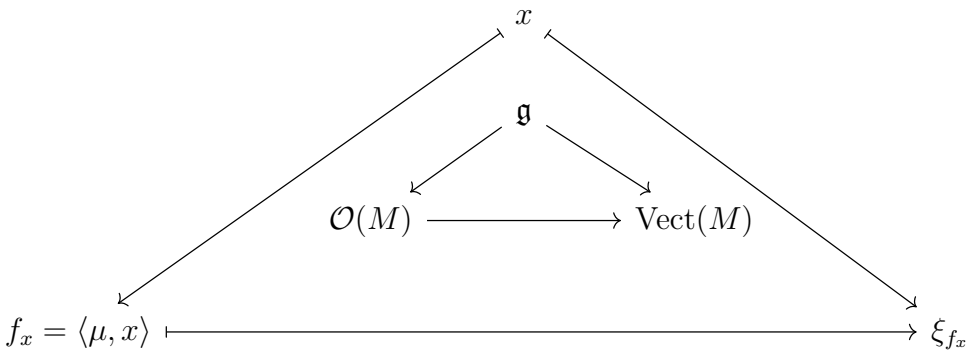
$$F = \cup F_{\mathcal{O}_{e'} \cap \mathfrak{n}_o}$$

*with each piece has maximal dimension. Namely each piece is a Lagrangian in* $\mathcal{O}_e \times T^*V$

For my own leaerning purpose, I wish to include a proof on the coisotropic property of equivariant moment map I used.

**Proposition 11.7.** *Let* $(M, \omega)$ *be a symplectic manifold with Hamiltonian* $G$*-action. Suppose in addition that* $\mu : M \to \mathfrak{g}^*$ *is* $G$*-equivariant, then:* $\mu^{-1}(0)$ *is either coisotropic or empty.*

*Proof.* Recall the diagram for moment map:




And the defining property of $\xi_{f_x}$ is that:

$$\omega(\eta, \xi_{f_x}) = \eta(f_x)$$

for any vector field $\eta$ on $M$. Therefore:

$$\omega(\eta, x) = \eta(f_x) = \langle \eta, d\langle \mu, x \rangle \rangle$$

Look at the expression at $p \in M$:

$$\omega(\eta_p, x_p) = (d\mu_p)(\eta_p)(x)$$

In other words, for any $v \in T_pM$ we have:

$$d\mu_p(v)(x) = \omega(v, x_p)$$

Now let $Z = \mu^{-1}(0)$, and $p \in Z$ be a smooth point, then:

$$T_pZ = \ker(d\mu_p) = \{v \in T_pM \mid \omega(v, x_p) = 0, \forall x \in \mathfrak{g}\} = T_p(G \cdot p)^{(\perp, \omega)}$$

Therefore it suffices to show that:

$$T_p(G \cdot p) \subset T_p(G \cdot p)^{\perp}$$

or in other word, $G \cdot p$ is isotropic. But for $x, y \in \mathfrak{g}$, we have:

$$\omega(x_p, y_p) = d\mu_p(y_p)(x_p)$$

But $\mu$ is constant along $G \cdot p$, so that $d\mu_p(y_p) = 0$ already. We're done. □

Finally, let's discuss the the number of irreducible components of the generalized Springer fibre:

$$\mathfrak{Y}_{e,e'} = G \times Z_{e,e'}$$

for $Z_{e,e'} = \{(x, v, p) \mid x + vp^t \in \mathcal{O}_e, x \in \mathcal{O}_{e'}\}$. If $e', e$ has partition type $\lambda, \mu$, the goal is to define a map:

$$Z_{e,e'} \to \{C_\beta \mid \lambda/\beta = \mu/\beta \text{ is a horizontal strip}\}$$

To construct the image of $(x, v, p)$, we set $y = x + vp^t$. Consider the subspace $W = \langle v, xv, \cdots, \rangle$ of dimension $k$. Note $x|_W = (x + vp^t)|_W$ and they just form a single Jordan block (this is the "strip" that we desired for). Therefore both $x, x + vp^t$ descend to a map on $V/W$. Moreover, they're equal on $V/W$ and give a partition $\beta$. This is the partition we want.

Recall that the interpretation of Littlewood-Richardson coefficient in the setting of nilpotent orbit is the following. Fix $\alpha \vdash k, \beta \vdash n - k, \lambda \vdash n$, and consider the space:

$$\mathcal{S}^\lambda_{\alpha,\beta} = \{U \in \text{Gr } (k, V) \mid x(U) \subset U, x|_U \in \mathcal{O}_\alpha, x|_{V/U} \in \mathcal{O}_\beta\}$$

and $c^\lambda_{\alpha,\beta}$ is the number of irreducible components of maximal dimension. This proves that our map above is well-defined. To see this map is surjective, we pick:

$$x = \begin{pmatrix} J_{(k)} & X \\ & J_\beta \end{pmatrix} \in \mathcal{O}_\lambda, y = \begin{pmatrix} J_{(k)} & Y \\ & J_\beta \end{pmatrix} \in \mathcal{O}_\mu$$

We must show we can conjugate $x$ such that it $gxg^{-1} - y$ has rank 1. But Let's try to conjugate $x$ by $\begin{pmatrix} 1 & A \\ & 1 \end{pmatrix}$:

$$\begin{pmatrix} 1 & A \\ & 1 \end{pmatrix} \begin{pmatrix} J_{(k)} & X \\ & J_\beta \end{pmatrix} \begin{pmatrix} 1 & -A \\ & 1 \end{pmatrix} = \begin{pmatrix} J_{(k)} & -J_{(k)}A + X + AJ_\beta \\ & J_\beta \end{pmatrix}$$

Note that if we plug in $A = E_{ij}$ the elementary matrix, we see:

$$JA + AJ = E_{i-1,j} - \delta(\beta, i, j)E_{i,j+1}$$

for $\delta(\beta, i, j) = 0, 1$. In any case we can conjugate $X$ so that it eventually concentrate on the last row. Similar procedure can be done for $y$ and we're therefore done.

Now let:

$$Z^\beta_{\lambda,\mu} = \{(x, v, p) \in Z_{\lambda,\mu} \mid x|_{V/W(x,v)} \text{ has type } \beta\}$$

The goal is to show that this set is irreducible. The idea is to use the fact that $Z^\beta_{\lambda,\mu}$ has huge symmetry. Consider:

$$Z^\beta_{\lambda,\mu} \to \{(v, W) \mid V \times \text{Gr } (k, V), v \in W\}$$

sending $(x, v, p) \mapsto (v, \langle v, xv, \cdots \rangle)$. This map is $G$-equivariant and surjective, and the fibre is a partial flag variety (hence is irreducible). So it suffices to show that one fibre is irreducible. Let's fix a basis $e_1, \cdots, e_n$ for $V$. Now let's look at the fibre at:

$$(e_k, \langle e_1, e_2, \cdots, e_k \rangle)$$

This consists of:

$$\{(x = \begin{pmatrix} X & Y \\ & Z \end{pmatrix}, p) \mid e_k \text{ is a cyclic vector for } X, Z \in \mathcal{O}_\beta, x \in \mathcal{O}_\lambda, x + e_k p^t \in \mathcal{O}_\mu\}$$

Let $Q_k = \begin{pmatrix} * & \cdots & * & 0 \\ * & \cdots & * & 0 \\ * & \vdots & \vdots & \vdots \\ * & * & * & * \end{pmatrix}$ be the mirabolic subgroup of $\mathrm{GL}_k$. Note that our space above admit a $Q_k \times \mathrm{GL}_{n-k}$-action (this group fixes $(e_k, W)$), and therefore it suffices to show one orbit is irreducible. Namely, we need to show:

$$\{(\begin{pmatrix} J_k & A \\ & J_\beta \end{pmatrix}, p) \mid x \in \mathcal{O}_\lambda, x + e_k p^t \in \mathcal{O}_\mu\}$$

We acutally have more symmetry. Note that the unipotent group:

$$\begin{pmatrix} I_k & * \\ & I_{n-k} \end{pmatrix}$$

normalizes $Q_k \times \mathrm{GL}_{n-k}$, so that we quotient out this symmetry, and for this we may assume $A$ has entries concentrated in the last row. Therefore, the residual variety is isomorphic to:

$$X_\lambda \times X_\mu$$

where: $X_\lambda \coloneqq \left\{ A = \begin{pmatrix} 0 & \cdots & 0 \\ 0 & \vdots & 0 \\ v_1 & \cdots & v_{n-k} \end{pmatrix} \mid \begin{pmatrix} J_k & A \\ & J_\beta \end{pmatrix} \in \mathcal{O}_\lambda \right\}$. Therefore it suffices to $X_\lambda$ is irreducible. But this follows from a simple calculation which shows that the condition for $v_1, \cdots, v_{n-k}$ are precisely of form some of them are zero and others are nonzero.